

New Power System Security and Low Carbon Transformation

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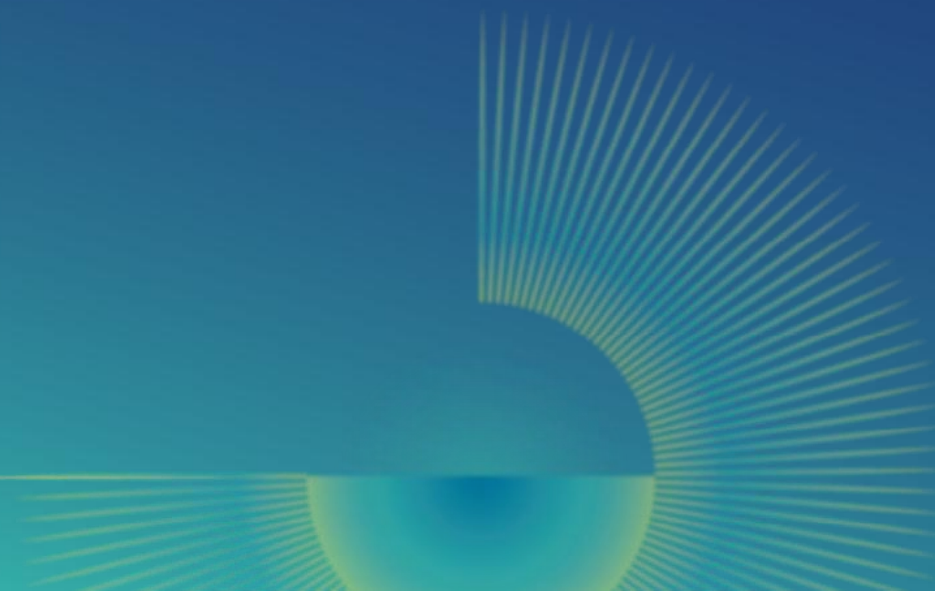
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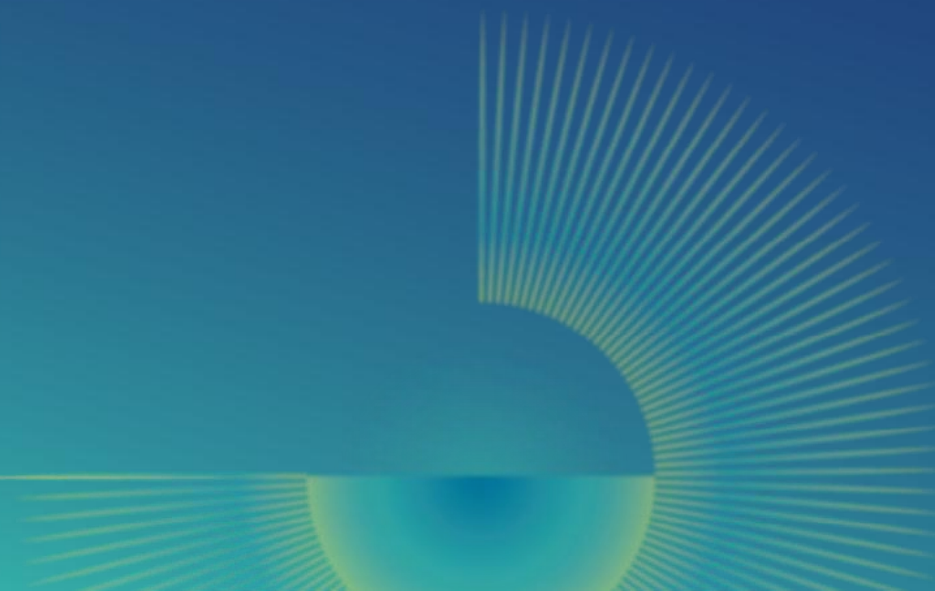
Electric Power University



New Power System Security and Low Carbon Transformation

新型电力系统安全与低碳转型

- New Power System Safety
新型电力系统安全
- Collaborative operation of source network load storage
wether and carbon
源-网-荷-储 与 气-碳 协同运行



ONE

CLEAN IN ACTION
GREEN
FOR
APEC

New Power System Safety



- **Efficient consumption and stable operation of a high proportion of new energy**
- **A large number of power electronic devices are widely integrated into the system**
- **Continuous load growth and user demand for high-quality power supply**
- **Conventional coal-fired power units are gradually reduced**

The operation of the new power system exhibits new characteristics

- 高比例新能源的高效消纳和稳定运行
- 大量电力电子装备广泛接入系统
- 负荷持续增长和高质量供电的用户需求
- 常规煤电机组逐步退出

新型电力系统的运行呈现新的特征

This presents new challenges for the safety of power systems
对电力系统的安全提出了新的挑战

◆ **System security**

◆ **Network and information security**

◆ **Personal safety**

◆ **Equipment safety**

◆ **Electrical safety**

◆ **Environmental safety**

System safety in new power systems

1 新型电力系统的系统安全问题

- ◆ The structure of power systems is becoming increasingly complex, it is urgent to improve the ability to balance the volatility and intermittency for the high proportion of renewable energy, and also to prevent system failures caused by natural disasters.

电力系统结构日益复杂，从系统角度平衡高比例新能源的波动性和间歇性、防范自然灾害导致系统故障的能力亟待提高。

- ◆ The increasing trend of power electronic devices conneted to the grid leads to a decline in system inertia, and also leads to abnormal changes in frequency response characteristics, and results in continuous weakening of the system's ability to balance source-load power fluctuations.

电力电子化趋势导致系统惯量水平下降，频率响应特性呈现异常变化，系统平衡源-荷功率波动的能力不断弱化。

- ◆ Load characteristics tend to be more constant and power-oriented, reducing the load's ability to self-regulate voltage, which increases the system's sensitivity to voltage fluctuations and challenges the ability to control voltage stability.

负荷特性更趋恒功率化，降低了负荷对电压的自调节能力，导致系统对电压波动的敏感性加大，电压稳定的控制能力受到挑战。

- ◆ The proportion of electricity received outside the urban city groups has increased year by year, while the reactive power support and voltage stability capacity of load centers have declined.

城市群外受电比例逐年增大，负荷中心的无功支撑和电压稳定能力下降。



System safety in new power systems

新型电力系统的系统安全问题

Large-scale power outages in the system are occurring frequently

系统大面积停电事故层出不穷

- ◆ On March 21, 2018, Brazil experienced a major blackout. Due to protection setting errors and the stability control system failing to activate, 21.73 million kilowatts of load was lost, severely affecting 14 states in northern and northeastern Brazil.
- ◆ On June 16, 2019, Argentina experienced a massive blackout. Due to the stability control system refusing to activate, almost the entire country experienced a blackout, affecting about 48 million people and parts of Chile, and Brazil adjacent power line to Argentina.
- ◆ On March 3, 2022, a large-scale power outage occurred in Taiwan's power grid, resulting in a loss of about 8.46 million kilowatts of load and about 5.49 million users, affecting almost every city in Taiwan. It became the most severe blackout in Taiwan's power grid since 1999.
- ◆ On August 9, 2019, the UK experienced a massive blackout. Due to the chain outages of multiple units such as offshore wind, distributed PVs, and gas-fired power, the outage affected most of England and Wales, affecting about one million users.
- ◆ On June 2024, Super Typhoon made a strong landfall along the coast of Hainan Province. Across the province, 79 35 kV substations were shut down, and 836 lines of 10 kV tripped. One-third of the province's users was affected to varying degrees.

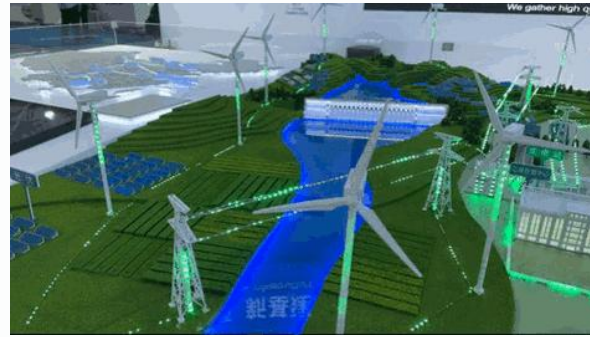


Solving power system security through intelligent means

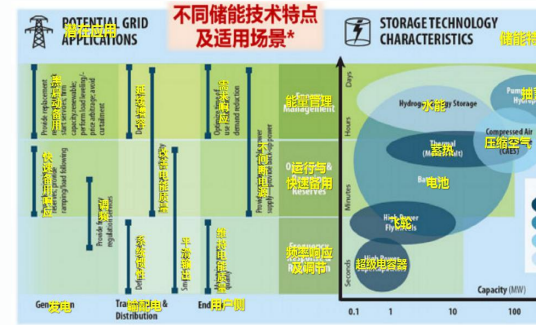
通过智能化手段解决电力系统 系统安全



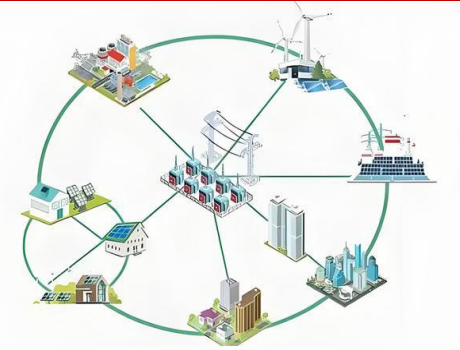
Intelligent dispatch of power systems
电力系统智能调度



Flexible dynamic adjustment of source-storage-load
源-储-荷柔性动态调节



safety measures of energy storage systems
安全的储能系统



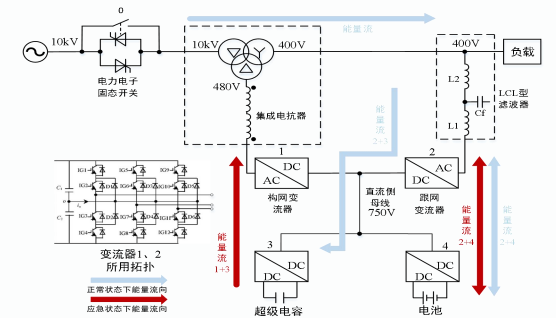
Rapid protection and self-healing of the grid
电网快速保护与自愈



Robots operate with electricity
机器人带电作业



Virtual power plant
虚拟电厂



Flexible interconnection of power distribution systems
配电系统柔性互联

Network and information security in new power systems

2 新型电力系统的网络与信息安全问题

Power grid information security will directly or indirectly affects the security of the entire power system. Information security threats may exist at every stage of generation, transmission, transformation, distribution, and usage. Once attacked, the consequences can be unimaginable. If control information is wrong, it will ultimately leads to widespread power outages and irreparable losses in the power system.

电网信息安全会直接或间接影响整个电力系统的安全，电力系统在发、输、变、配、用各环节都可能存在信息安全威胁，一旦受到攻击，后果不堪想象。如果控制信息出错，可能造成电力系统大面积停电和难以挽回的损失。

- On December 17, 2016, hackers attacked Ukraine's national power sector, causing a massive blackout lasting 30 minutes.
- On November 7, 2021, DMEA, a power company in Colorado, USA, suffered a ransomware attack, causing disruptions to phone, email, billing, and customer account systems.
- On March 21, 2018, an Indian power company was hacked to steal customers' bill data and extort 10 million rupees.

Venezuela power outage incident

委内瑞拉停电事件

- ◆ 2019年3月7日傍晚（当地时间）开始，委内瑞拉国内包括首都加拉加斯在内的大部分地区停电超过24小时，在委内瑞拉23个州中，一度有20个州全面停电
- ◆ 停电导致加拉加斯地铁无法运行，造成大规模交通拥堵，学校、医院、工厂、机场等都受到严重影响，手机和网络也无法正常使用。
- ◆ 8日凌晨，加拉加斯部分地区开始恢复供电，随后其他地区电力供应也逐步恢复，但是9日中午、10日的再次停电，给人们带来巨大恐慌。

- On March 7, 2019, Venezuela's power grid was hit by a cyberattack, causing widespread blackouts in 20 states nationwide, with some areas experiencing outages for 24 hours.



Network and information security in new power systems

新型电力系统的网络与信息安全问题

External information risks and threats

外部信息风险威胁

- ◆ **Hacker attacks:** Hackers infiltrate power systems through networks, disrupting system stability or stealing critical information.
- ◆ **Virus infection:** Malware infects power systems through various channels, causing system crashes or data leaks.
- ◆ **Social engineering attacks:** Attackers use social tools or identity disguises to deceive or entice confidential employees to obtain critical information.

Internal information risk threats

内部信息风险威胁

- ◆ **Employee misoperation:** Employees accidentally delete or modify key system data, causing system failures.
- ◆ **Internal attacks:** Criminals use their authority to sabotage or steal sensitive company information.

Optimized operation of power and information systems in coordination

电力与信息系统协同优化运行

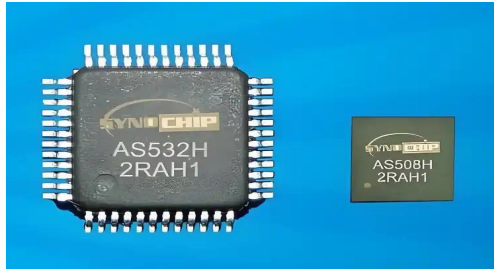
The power system integrates with information systems, supporting the coordinated optimization of source, grid, load, and storage through technological innovation, promoting the safe and stable operation of the power system.

Strengthen cybersecurity measures to ensure the security of the power system's network and information

加强网络安全防范，确保电力系统网络与信息安全

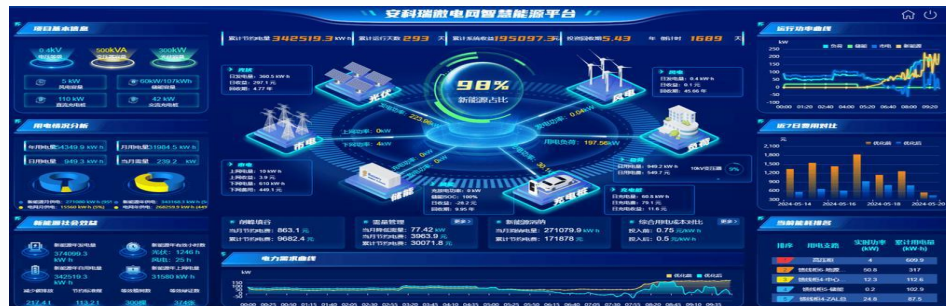
Develop specialized security chips

研制专用安全芯片



Strengthen information security offense and defense drills

加强信息安全攻防演练



Power information network attack and defense range system
电力信息网络攻防靶场系统

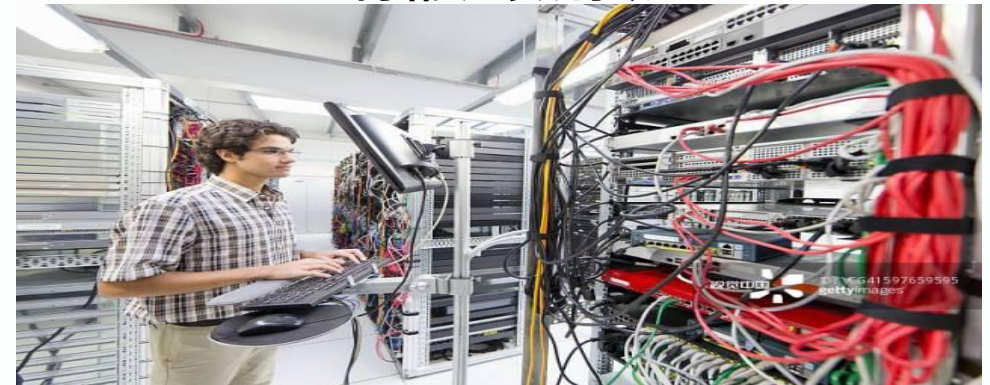
Secondary system cascade information security protection

二次系统梯级信息安全防护体系



Network Security Training System

网络安全实训系统



Personal safety issues in new power systems

3 新型电力系统的 人身安全问题



**Tower collapse
fatal accident**

倒塔死亡事故



**Equipment
hazard accidents**

设备隐患事故



**Accident caused by electric
leakage causing fatal injury**

漏电击中身亡人身事故



**High-voltage operation
electric shock accidents**
高压操作触电事故



Explosion and fire accident
电力爆燃火灾事故

Report of the 20th National Congress of the CPC

Major and especially serious emergencies, whether natural disasters or liability accidents, all involve varying degrees of issues such as failure to implement primary responsibilities, incomplete hazard investigation and management, incomplete laws and standards, lax safety supervision and law enforcement, incomplete regulatory systems and mechanisms, weak safety foundations, and weak emergency rescue capabilities.

Ensuring personal safety in the power system through management innovation

通过管理创新保障电力系统 人身安全

◆ Strengthen safety supervision for personnel in new power systems 加强新型电力系统人员的安全监管水平

- Shifting from post-event handling to proactive prevention
- Supervision measures shifted from rigid to flexible regulation
- Investment shifted from administrative bottom-line protection to cost reduction and efficiency improvement
- Safety supervision responsibilities shifted from being vague and unpredictable to clearly determined

◆ Innovate technical means for safety supervision 创新安全监管的技术手段

IoT technology + precise algorithms +
technical monitoring

=

Full-process safety supervision for front,
middle, and back ends of the project

物联网技术+精确算法+技术监控

=

项目的前、中、后端全程安全监管



A safety operation management system based on Alot high-precision positioning and image analysis — the safe electricity management cloud platform adopts a layered distributed structure design. When an anomaly occurs on site, an alarm signal is pushed via the system platform or mobile app, and SMS notifications are sent to the safety responsible person, with operations staff dispatched to address on-site hazards.

Ensuring personal safety in the power system through management innovation

通过管理创新保障电力系统 人身安全

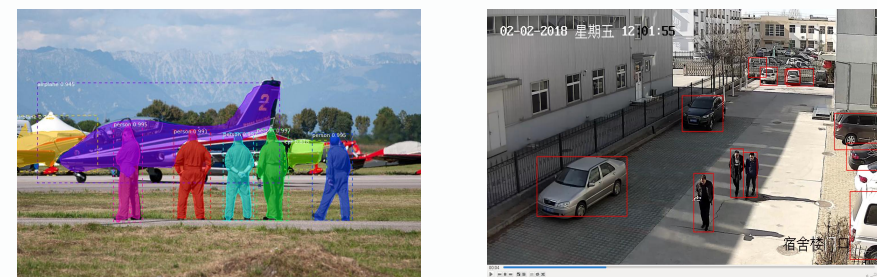
Traditional manual inspections are upgraded to an automated, visualized, and intelligent operation and maintenance mode, by utilizing technologies such as AI, the Internet of Things, digital twins, and augmented reality (AR), .



Intelligent inspection and visual inspection platform
智能巡检与可视化巡检平台

Through frequent and comprehensive training and drills, the personnel skill level of power system operation and maintenance is enhanced.

通过高频度、全方位的培训演练，提升电力系统运维人员技能水平。



Video image recognition and intelligent monitoring applications

视频图像识别及智能监控应用

Equipment safety issues in new power systems

4 新型电力系统的设备安全问题



Fire accident at Tianshan Converter Station



Fire and explosion accident at the 330KV substation in southern Shaanxi



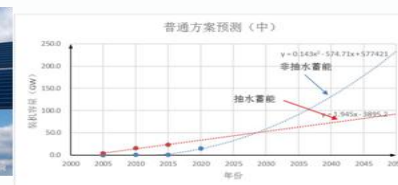
Fire accident at chemical battery charging facilities



Equipment damage caused by the earthquake



Equipment accidents during installation, commissioning, and operation for offshore wind power
海上风电在安装、调试、运行中的设备事故



Energy storage, hydrogen storage, and hydrogen production systems will develop on a large scale, and equipment safety issues will attract widespread attention.

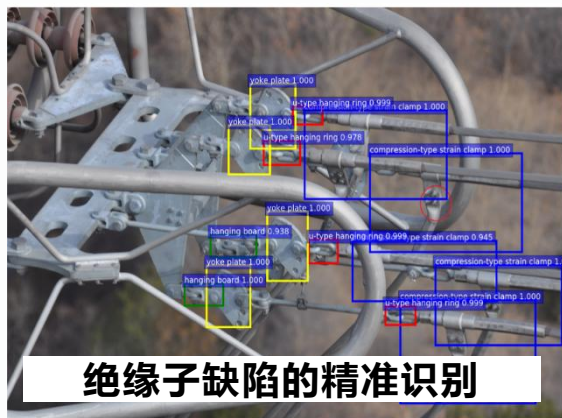
储能电站、储氢电站、制氢系统将规模化发展，设备安全问题引起广泛关注

Equipment damage accidents occur frequently, for example, fires and explosions of energy storage batteries, high-voltage cables and transformers, GIS, and other equipment. Natural disasters like earthquakes and severe weather will also occasionally cause the damage of equipment.

储能电池、高压电缆、变压器、GIS等设备起火爆炸、地震及恶劣气候等自然灾害导致的设备损坏事故时有发生。

Maintaining the safety of power system equipment through technological innovation

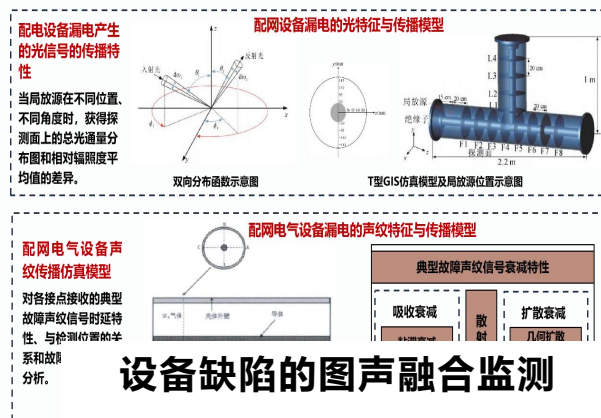
通过技术创新维护电力系统设备安全



Accurate identification of insulator defects



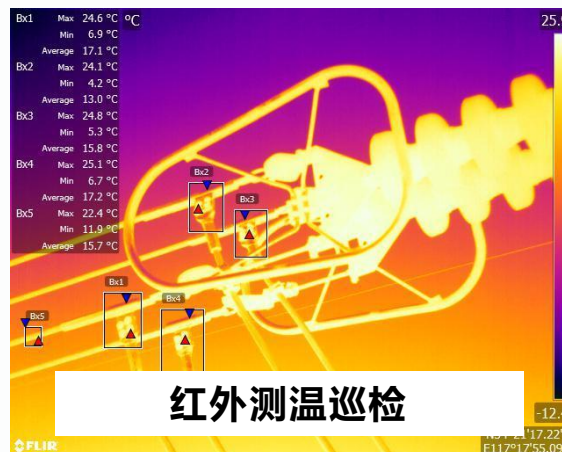
Joint air-ground monitor of transmission lines



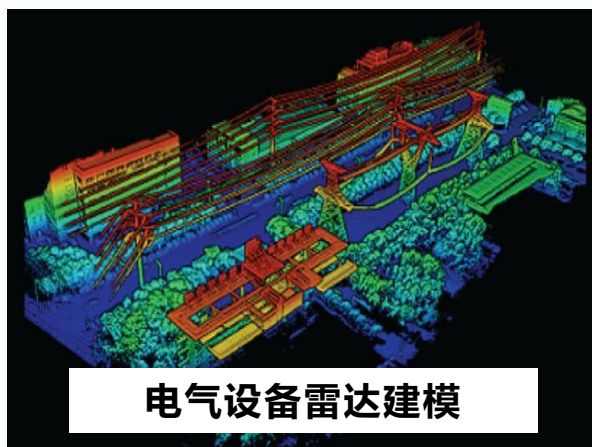
Audio-visual fusion monitoring of equipment defects



Intelligent monitoring of substations



Infrared temperature measurement inspection



Radar modeling of electrical equipment



Cable fault diagnosis



Online analysis of sulfur hexafluoride equipment faults

LV system safety issues in new power systems

5 新型电力系统的 用电安全问题

- ◆ The widespread adoption of distributed PV has led to local voltage increases in distribution and power supply systems, causing transformer overloads, which not only imposes higher demands on grid operation control but also introduces new challenges to electricity safety.
 - ◆ Power quality issues such as voltage dips and harmonics can cause production stop or damage to sensitive equipment.
 - ◆ Safe operation and leakage protection for low-voltage electrical equipment become more serious.
-
- ◆ 分布式光伏大量接入，导致配用电系统局部电压升高，变压器过载，不仅对电网运行控制提出更高要求，也对用电安全提出新的挑战。
 - ◆ 电压暂降、谐波等电能质量问题造成敏感设备停产或损坏。
 - ◆ 低压电器设备的安全运行和漏电保护

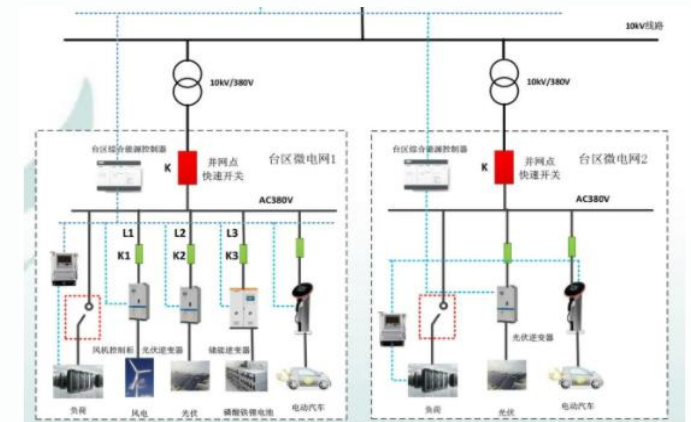
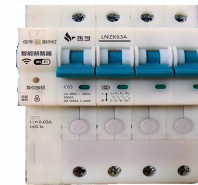


Strengthen publicity, enhance ubiquitous sensing and informatization, and use innovative means to solve LV system safety issues

加强宣传，增强泛在感知和信息化等创新手段解决用电安全问题

- ◆ Strengthen publicity and education on electricity safety for all citizens.
- ◆ Developing cost-effective and practical sensor chips, shifting the advanced sensing, big data interconnection, and information technology from the power grid to low-voltage power networks.
- ◆ Methods such as DVR and active filter are used to improve power quality.
- ◆ Improve the emergency response capabilities for enterprises.

- ◆ 加强全民用用电安全的宣传教育。
- ◆ 研发价低实用的传感芯片，将电网先进的泛在感知、大数据互联和信息化技术下沉应用于低压用电网络。
- ◆ 应用DVR、有源滤波等手段提高电能质量水平。
- ◆ 提高用电企业的应急能力和响应水平。

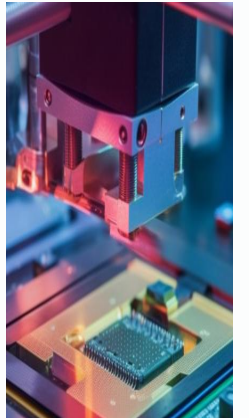
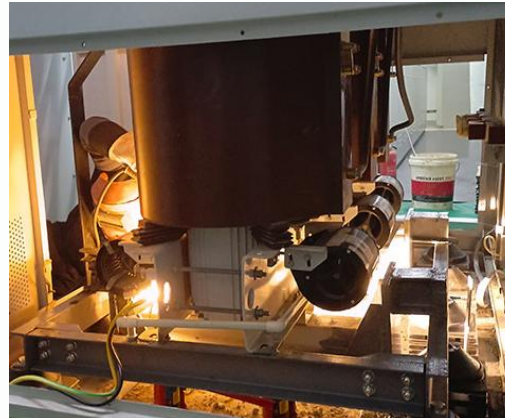
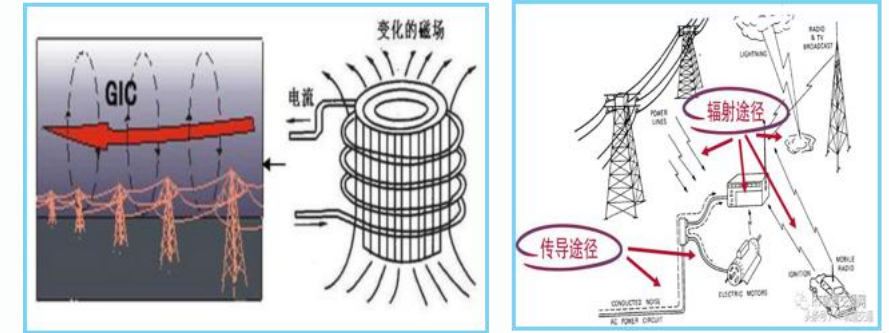


Environmental safety issues of new power systems

6 新型电力系统的环境安全问题

- Research and development the environmental monitoring chips
- Research and development the new environmentally friendly insulation materials
- Electromagnetic environment monitoring and management
- Monitoring and control of transformer noise
- Monitoring and control of construction site dust
- Research the impact of space weather and geomagnetic storms on transmission lines
- Sulfur hexafluoride gas leak monitoring
- Electromagnetic environment risk early warning and control

- 环境监测芯片研发
- 新型环保绝缘材料研发
- 电磁环境监测及治理
- 变压器噪声监测及治理
- 工地扬尘监测及治理
- 空间天气及地磁暴对输电线路的影响
- 六氟化硫气体监测
- 电磁环境风险预警及控制



- ◆ On February 29, 2024, the CPC Central Committee held its collective study session on new energy technology and China's energy security.
- ◆ Professor Liu Jizhen gave an explanation to the colleagues of the Central Political Bureau and offered work suggestions.

- ◆ 2024年2月29日下午，中共中央政治局就**新能源技术与能源安全**进行**第十二次集体学习**。
- ◆ 我校**刘吉臻教授**为中央政治局的同志们进行讲解，并提出工作建议。

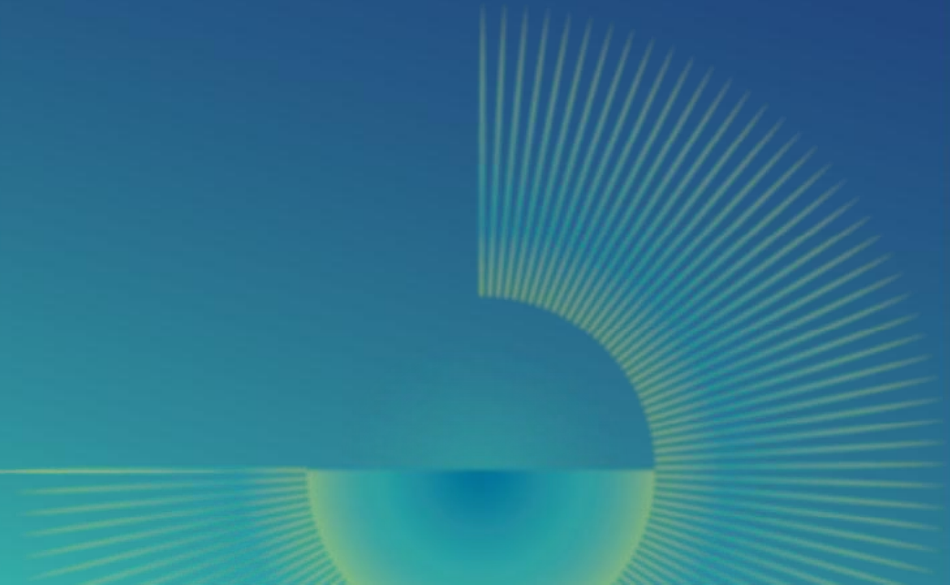
Energy security is closely tied to the overall development of the economy and society. We must make greater efforts to promote high-quality development of new energy and provide safe and reliable energy guarantees for Chinese-style modernization.



能源安全事关经济社会发展全局。我们要以更大力度推动新能源高质量发展，为中国式现代化建设提供安全可靠能源保障。

TWO

**Collaborative operation of source network
load storage w e t h e r and carbon**



Active Distribution Network

1 主动配电网

A fundamental operation mode has been changed in the traditional power distribution networks .

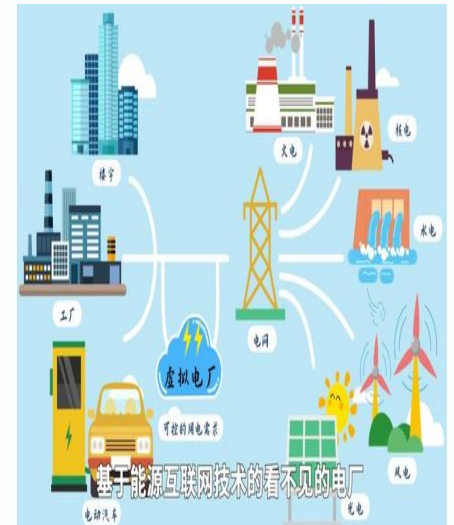
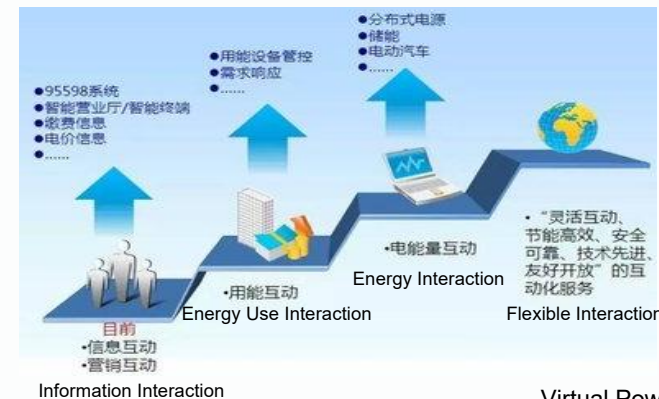
传统配电网运行形态发生了本质变化

- ◆ Large-scale integration of distributed power sources and energy storage has been interconnected in and changed the operation mode of traditional distribution networks.

分布式电源和分布式储能的大规模接入改变了传统配电网的运行形态

- Active distribution networks, microgrids, and virtual power plant based on energy internet are driving traditional distribution networks transfer from traditional into new distribution systems.

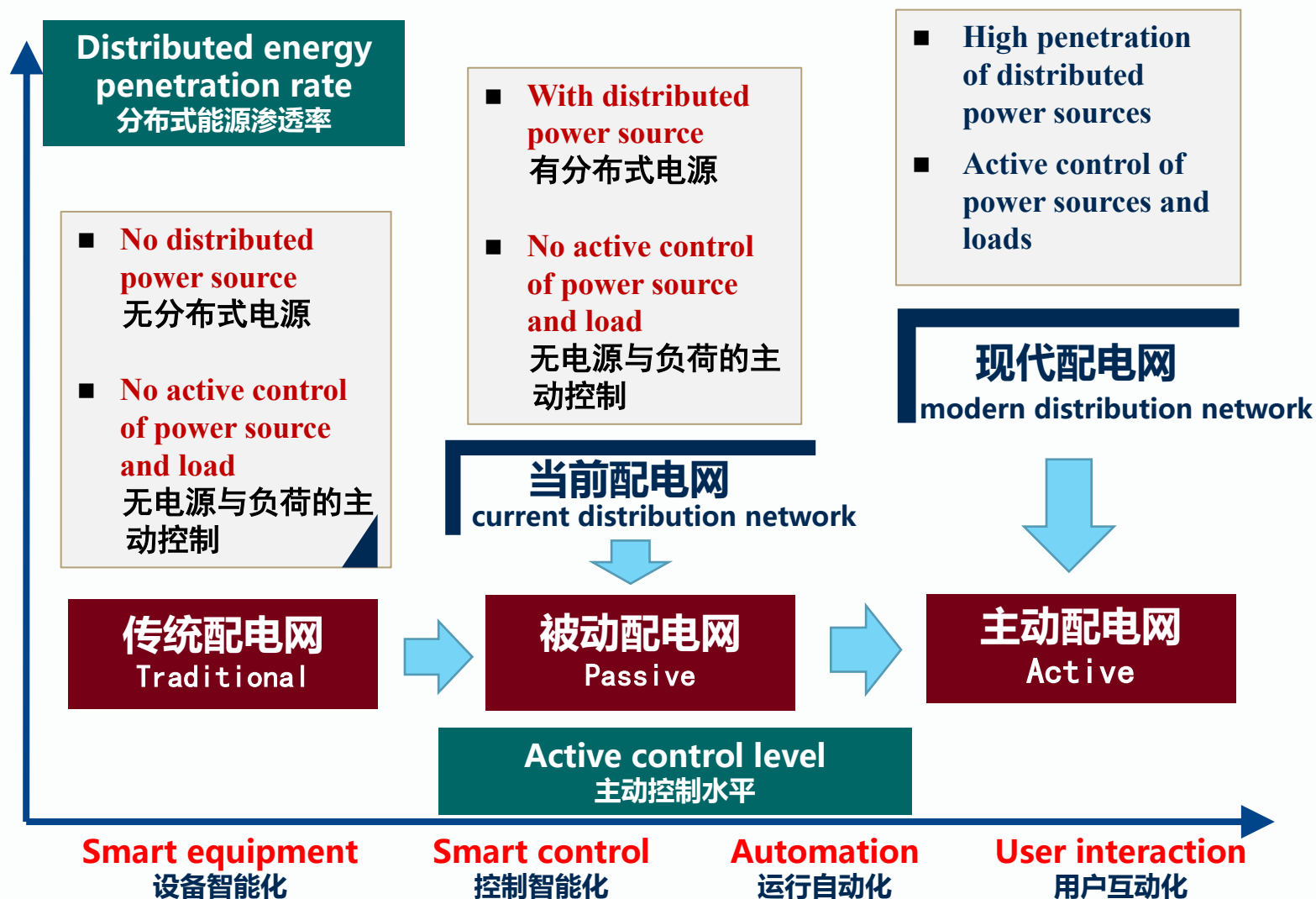
主动配电网、微电网和基于能源互联网的虚拟电厂技术推动传统配电网向新型配电系统转型



Virtual Power Plant (VPP) Based on Energy Internet Technology

Distributed energy has driven the transformation of the distribution network

分布式能源推动了配电网的形态转变



Management of active distribution networks

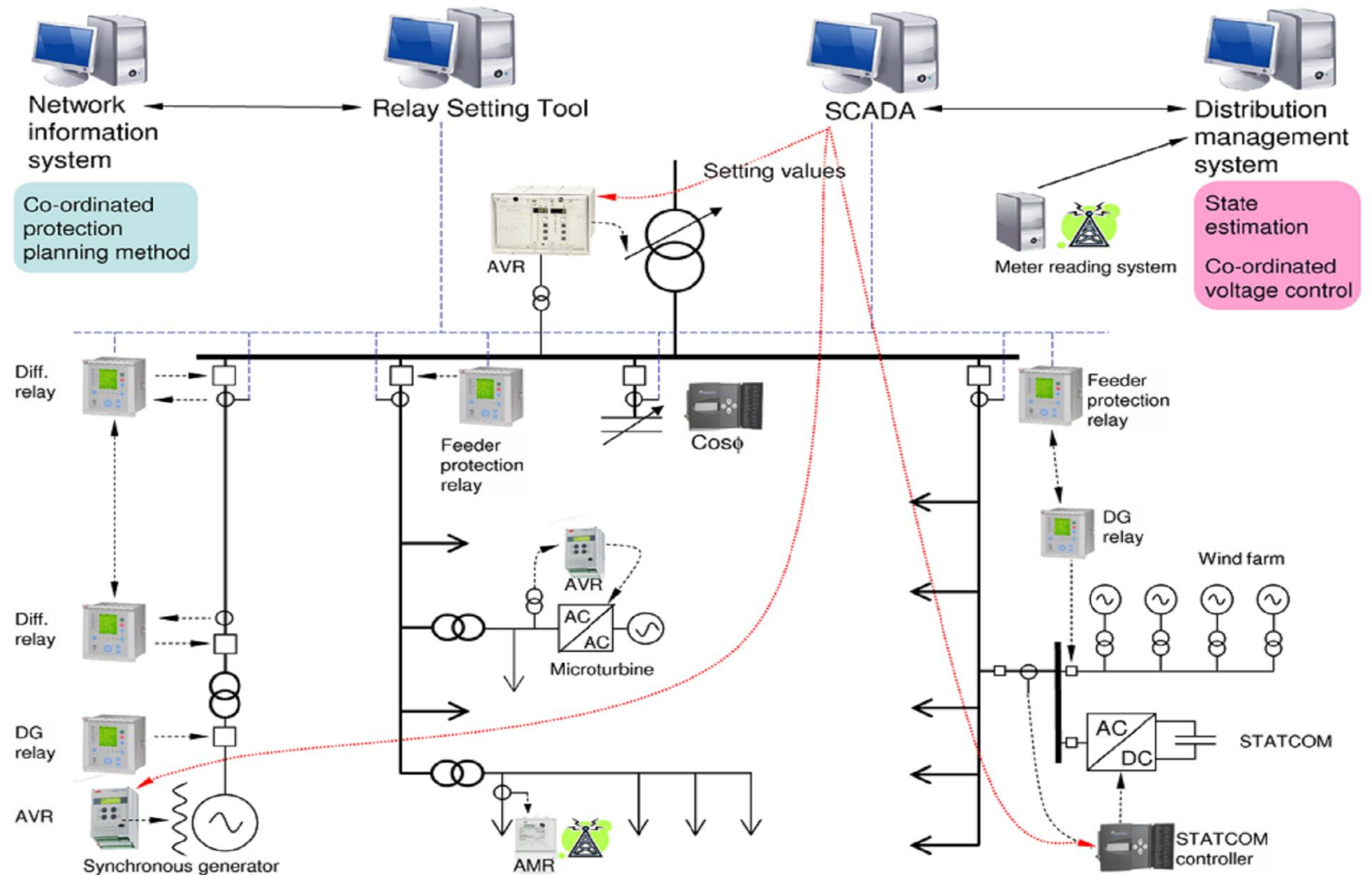
主动配电网的管理

All hardware devices such as

- relay protection,
- tap changers,
- automatic voltage regulators of distributed sources,
- STATCOMs,
- power factor controllers

are under decentralized, while above the decentralized control, there is still a centralized control system for distribution management.

所有硬件装置诸如继电保护、分接头和分布式电源的自动电压调节器、STATCOM和功率因数控制器等均为分散控制，在分散控制的上层仍有配电管理的集中控制系统。

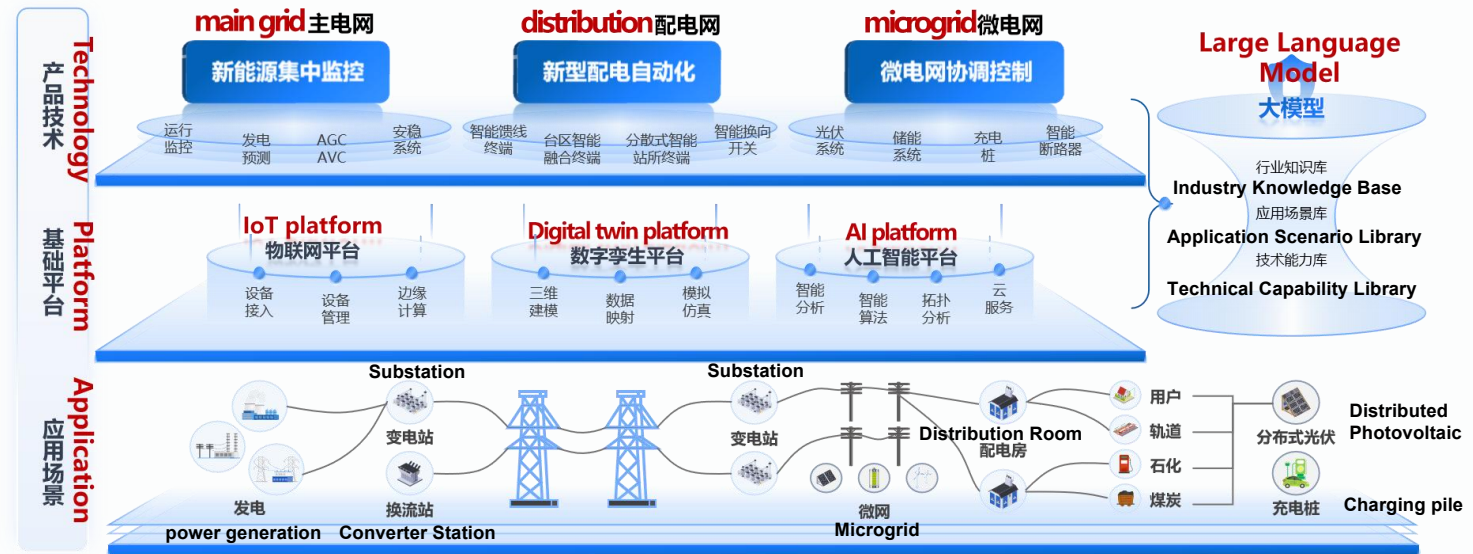


Coordinated operation and control of main grid, distribution net and microgrid

2 主配微网协同运行控制

Integrate the "main-distribution-micro" power grids to provide guarantees for the coordinated development of power systems

贯通“主-配-微”电网，为主配微网协同发展提供核心保障



For main grid, we have to build an integrated centralized monitoring system for new energy stations to achieve "operational monitoring - intelligent operation and maintenance - operational decision-making", providing full-link support control to smart operation for centralized renewable energy stations, and comprehensively enhancing the operational efficiency and market competitiveness of power stations.

构建新能源场站一体化集中监控系统,实现“运行监控-智能运维-运营决策”，为新能源场站提供从设备控制到智慧运营的全链路支持，全面提升电站运营效益与市场竞争力。

Coordinated control of main, distribution and microgrid—New distribution automation system

主配微网协同运行控制 - 新型配电自动化系统

For distribution grid, regional distribution networks has to be transparently monitored, to achieve active balance control for source-grid-load-storage.

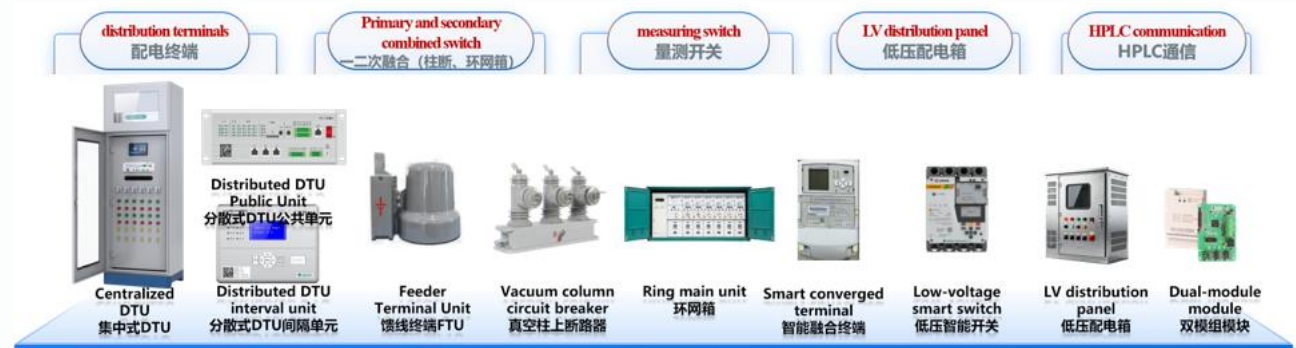
The new distribution automation system is equipped with various data acquisition, operation and control capabilities, enabling transparent perception, fault self-healing, active balancing, computational deduction, and new energy planning.

区域配电网透明化监控、源网荷储主动平衡控制。

新型配电自动化系统具备各类数据采集与处理、操作与控制等，实现透明感知、故障自愈、主动平衡、计算推演、新能源规划等。

Based on various types of equipment such as primary and secondary integrated terminals, develop a full product line of smart distribution terminals based on independently controllable technologies.

依托一二次融合终端等多类设备，研发基于自主可控技术的全产品线配电智能终端





Coordinated control of main, distribution and microgrid—Microgrid coordinated control system

主配微网协同运行控制 - 微电网协调控制系统

For microgrid, ensure the balanced, stable, efficient, and economical operation of microgrids and to promote the consumption and utilization of distributed new energy.

Using intelligent microgrid coordinated devices and multi-energy monitoring and execution terminals, such as PV, energy storage, and charging control systems, as well as intelligent load devices or systems to achieve the coordinated and optimized operation.

微电网平衡稳定并高效经济地运行，促进分布式新能源的消纳利用。应用智能微网协调控制装置、多种能源监控执行终端，集成光储充等新能源装备、智能负荷设备或系统，实现分布式电源、储能和负荷的协调优化运行及安全稳定控制。

For distribution transformer area, making cloud-edge collaboration and active balance, focusing on source-grid-load-storage resources in distribution transformer areas.

Construct a LV distribution network data decision center, equipped with intelligent terminal devices, carry out advanced applications, and gradually achieve power supply automation, transparency of PV-storage-charging, and digital intelligence distribution-microgrid coordination in distribution transformer areas.

云边协同、主动平衡，聚焦配变台区源网荷储资源。构建低压配网数据及决策中心。配套智能终端设备，围绕“融合终端+”开展高级应用，逐步实现配变台区的供电自动化、光储充透明化、配微协同数智化。

Coordinated control of main, distribution and microgrid—Intelligent Operation and Maintenance System for Power Distribution Switchgear

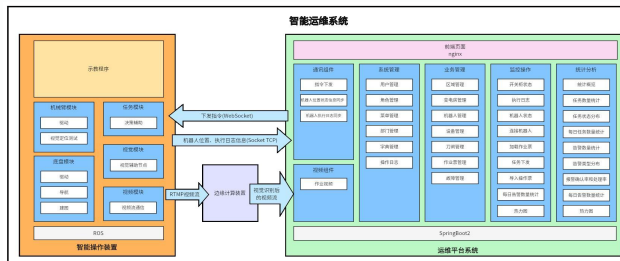
主配微网协同运行控制 - 配电开关柜智能运维系统

Intelligent sensing and fault detection of MV distribution switchgear
中压配电开关柜智能感知与缺陷检测

Operating space for power distribution switchgear
配电开关柜操作空间



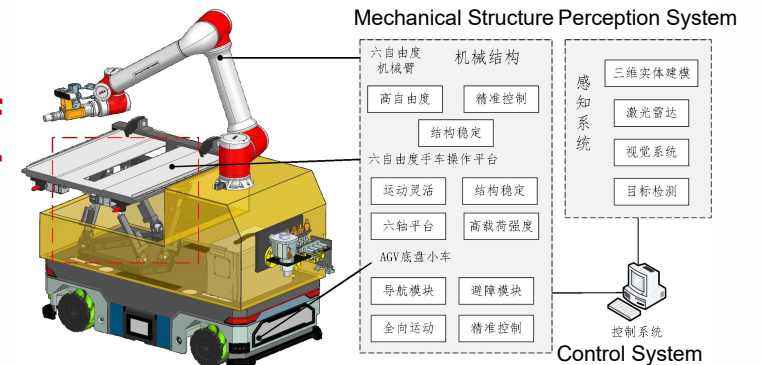
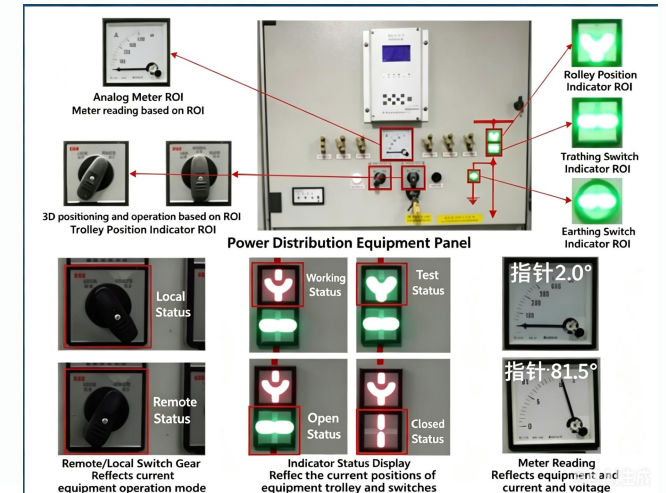
Autonomous operation and maintenance system for power distribution switchgear
配电开关柜设备自主运维检修系统



Intelligent operation, maintenance, and inspection of power distribution switchgear by a six-degree-of-freedom robotic arm

六自由度机械臂配电开关柜智能运检操作

Autonomous operation for power distribution switchgear
配电开关柜精细操作自主作业



Fast protection and self-healing of distribution networks

3 配电网快速保护与自愈

Synchronous measurement and feature extraction of fault transient signals in power electronics-based networks

电力电子化配电网故障暂态信号同步测量与特征提取

With the large-scale integration of distributed PVs into the distribution network, the fault transient signals have changed. So it is needed to investigate the influence mechanism of the distribution topology on fault transient signals, as well as the weak signal characteristics of high-impedance grounding faults.

大量分布式光伏接入配电网，故障暂态信号发生了变化。研究配电系统拓扑结构对故障暂态信号的影响机理，和高阻接地故障的微弱信号特征。

Precise location of grounding fault in power electronics-based distribution network

电力电子化配电网接地故障位置精确定位

Under the grid-connection of distributed PV, it is proposed to take into consideration of the impact of harmonics on neutral grounding devices, development mechanism of harmonic overvoltage, the causes of fault arcition, single-phase grounding fault handling method.

分布式新能源并网下，谐波对中性点接地装置的影响，谐波过电压的发展机理，故障电弧重燃的原因，提出配电网单相接地故障处理方法。

Fast protection and self-healing of distribution networks

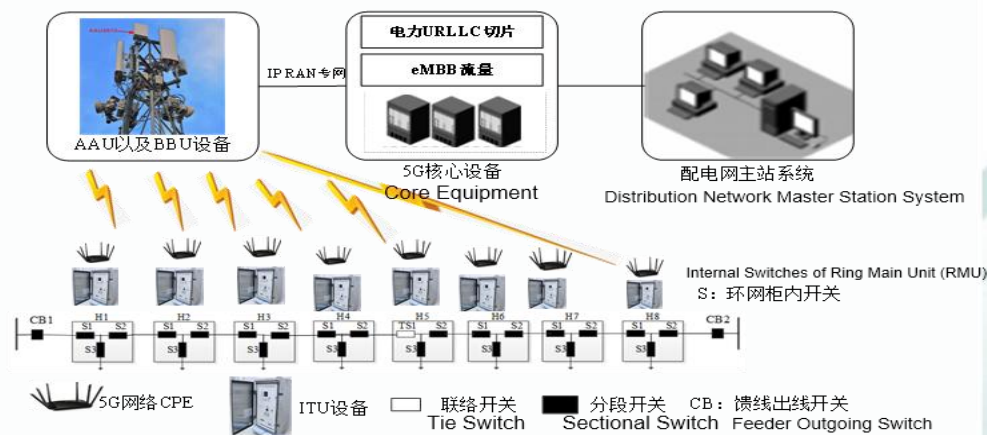
配电网快速保护与自愈

In non-fault power-off areas, load transfer is carried out based on new-generation communication technology

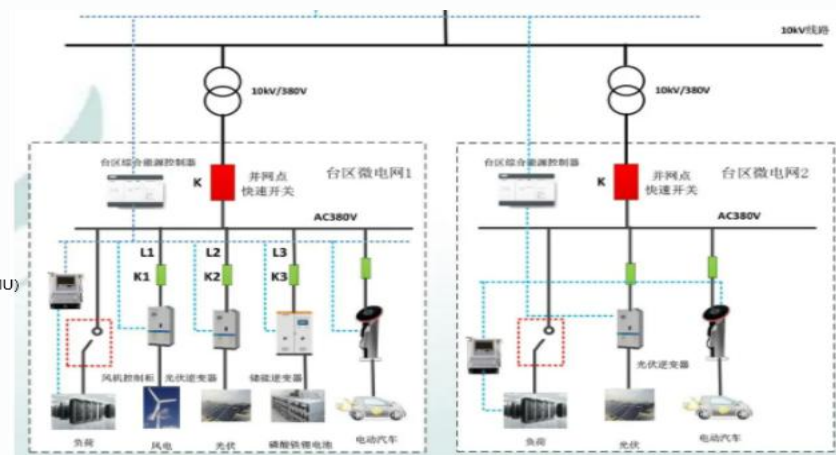
基于新一代通信技术的非故障失电区的负荷转供

Taking the new intelligent power distribution terminal as the distributed agent, fast identification, fast isolation, and power transfer technologies for grounding faults based on intelligent algorithms are studied

以新型智能配电终端为分布代理，研究基于智能算法的接地故障快速辨识，接地故障快速隔离与转供技术。



Load transfer in non-fault power-off areas based on 5G
基于5G实现非故障失电区的负荷转供示意图



Active islanding recovery in extreme cases
极端情况的主动孤岛恢复

Fast protection and self-healing of distribution networks

配电网快速保护与自愈

New type of power distribution terminal

新型配电终端

The new power distribution terminals include Beidou synchronous timing, chip-based protection, fault perception, and 5G communication, enabling rapid faults location, isolation of different types, as well as fast and optimal fault recovery.

新型配电终端具备**北斗同步授时**、**芯片化保护**、**故障全息感知**、**5G通信功能**，实现不同类型故障的快速、精确定位与隔离，以及快速最优化的故障恢复。



Safe operation of the secondary system of the distribution network

配电网二次系统的安全运行

The power distribution secondary system ensures the safe operation of the power grid and is a complex cyber-physical system with deep integration of equipment and data.

- ◆ Distribution integrated with edge security agent components achieve 100% control over the operational behaviors during the startup and running processes of terminal devices.
- ◆ The plug-in design of a dynamically computing environment solves the protection problem against complex and hidden attacks, such as illegal operations on the master station.

配电网二次系统保障电网安全运行，是设备-数据深度融合的复杂信息物理系统。

- 集成边缘安全智能体组件的配电终端，实现终端设备启动与运行过程操作行为的100%管控。
- 动态可信的计算环境插件化设计，解决主站遭非法操作等复杂隐蔽攻击的防护问题。

Intelligent monitoring of substations and lines

4 变电站及线路智能监控

Apply electrical inspection robots, intelligent inspection terminal equipment, etc., to achieve reduced-manpower inspection and intelligent monitoring of power grid, make standardized processes to realize the safe and standardized operation of maintenance, inspection, and control.



应用电气巡检机器人，智能巡检终端设备等，实现电网少人巡检、智能监控，制定标准化、可视化流程，实现检修、巡检和操控的安全化和规范化操作。

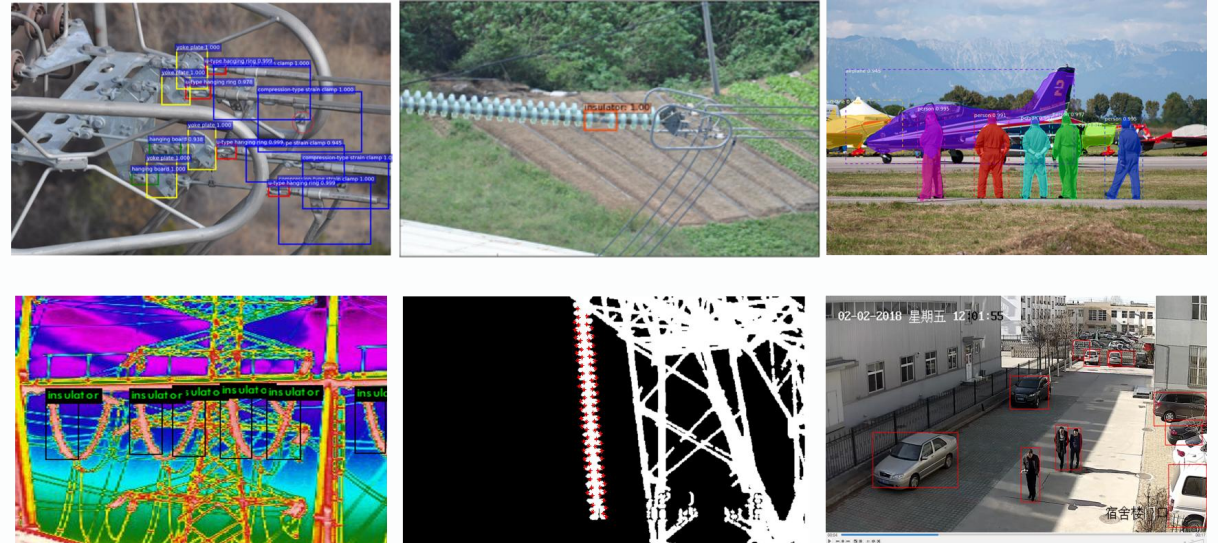
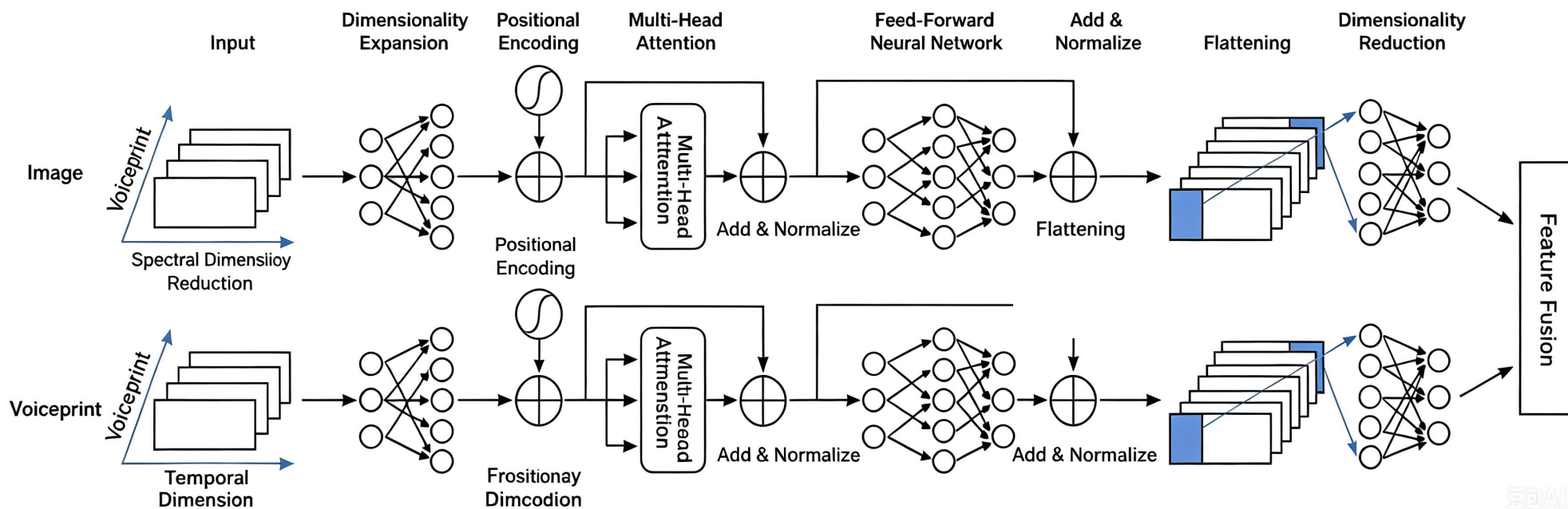


Image and voiceprint fusion monitoring of electrical equipment defects

电气设备缺陷的图像声纹融合监测

It will support image and voiceprint recognition of defects for equipment, achieves real-time monitoring, and provides accurate equipment diagnosis and fault early warning.

支持配网电气设备缺陷的图像识别和声纹识别，通过融合监测实现对配网设备的实时监控，提供精准的设备状态诊断和故障预警。



Line loss reduction for low-voltage transformer area

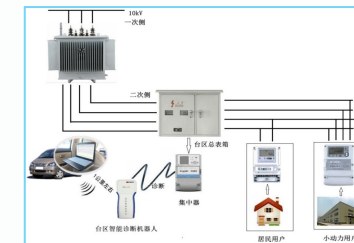
低压台区线损治理

Regarding the excessive line loss in the transformer area, analyze the main causes and provide a recommendation for rapid diagnosis and treatment, so as to make the loss rate of the transformer area reaches the required qualified index range.

针对台区线损过高问题，分析主要原因并出具整改方案及建议，快速诊断治理，使台区同期线损率达到要求的合格指标范围。

Within the same power supply area, an intelligent transformer district diagnosis robot is installed at each transformer district concentrator to perform data diagnosis.

在相同供电区域内，所有台区集中器处分别安装台区智能诊断机器人，进行数据诊断。

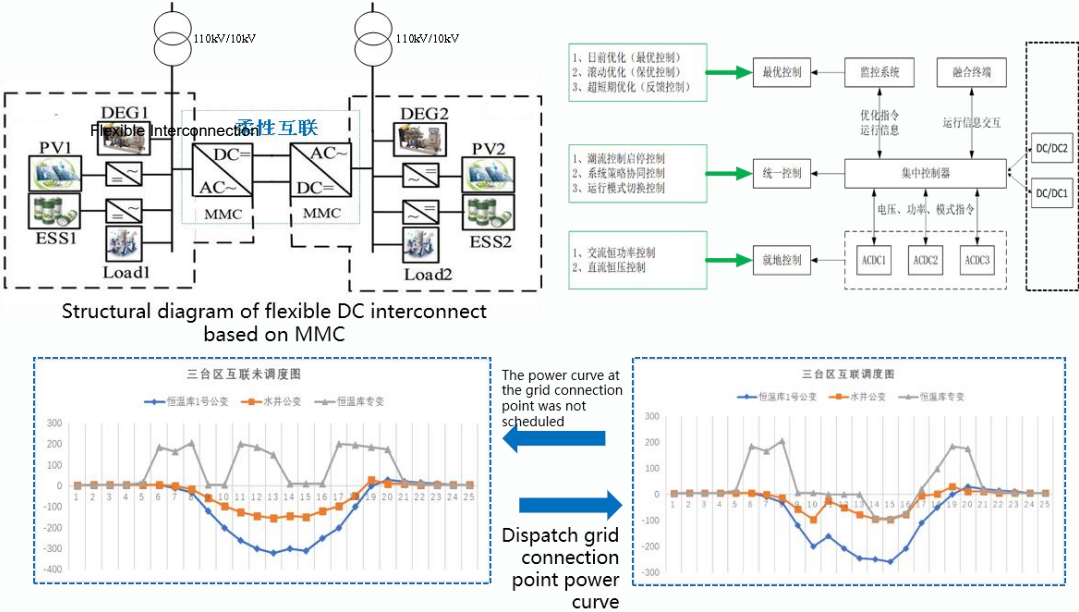


Flexible DC interconnection in the distribution power station

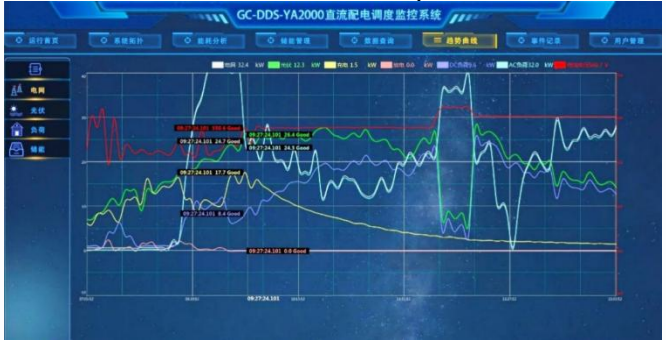
5 配电台区的柔性直流互联

Through flexible DC interconnection, mutual power exchange between transformer areas is achieved, providing power support for the AC distribution network during instantaneous ground faults. The interconnection system works as a virtual energy storage, allowing distributed PVs, energy storage, and DC loads to be directly connected to DC busbars, improving the reliability and flexibility of urban distribution systems.

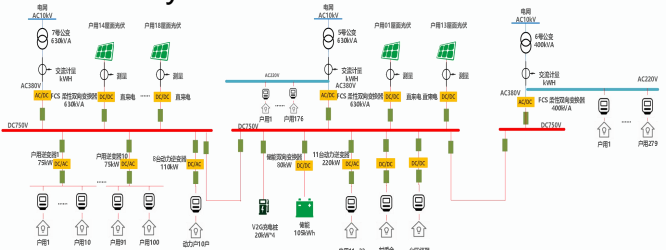
通过柔性直流互联，实现台区之间的功率互济，在瞬时接地故障时为交流配网提供功率支撑，互联系统具有虚拟储能特征，分布式光伏、储能和直流负荷可以直接接入直流母线，提高了城市配电系统的可靠性和灵活性。



CNOOC's zero-carbon park flexible DC interconnection system



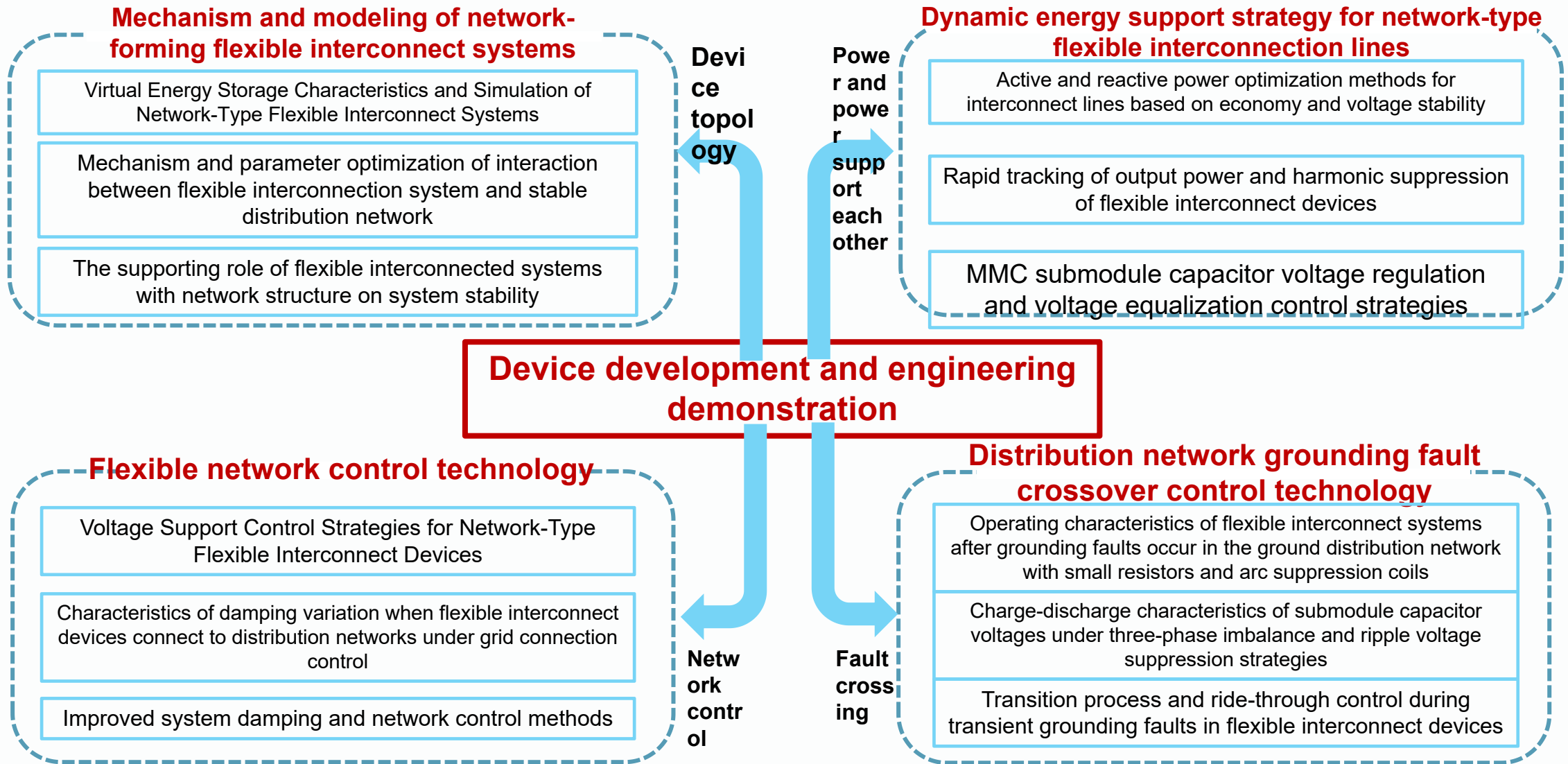
Nanjing Guochen Office Building AC-DC Hybrid Distribution Network



Nantou New Village in Linyi, Shandong, with three ends interconnected with flexible and straight connections

DC flexible interconnect system technology based on network control

基于构网控制的直流柔性互联系统技术





Pay attention to the technological development of AI

关注人工智能的技术发展

AI - A Powerful Engine for Product R&D

- ◆ Unmanned aerial vehicle (UAV) and robot vision:
- ◆ Edge computing with AI algorithms for end-monitoring devices:
- ◆ Application of machine tactile chips in fire monitoring of energy storage power
- ◆ Intelligent identification of defects in transmission and distribution lines:
- ◆ Image and acoustic fusion monitoring for equipment defects:
- ◆ Power environment monitoring:
- ◆ Others

人工智能 - 产品研发的强劲引擎

- 无人机和机器人视觉:
- 末端监测装置的边缘计算AI算法:
- 机器触觉芯片在电化学储能电站火灾监测中的应用
- 输配电线路金具缺陷智能识别:
- 电气设备缺陷的图像声纹融合监测:
- 电力环境监测（噪声/SF6泄露/扬尘/电磁污染）:
- 其他



Pay attention to the technological development of AI

关注人工智能的技术发展

AI-based security and prevention technology

基于人工智能的安全防控技术

- Intelligent robot inspection for substations, multimedia monitoring for distribution rooms, smart safety supervision system
- Health status detection and risk early warning of on-load changers for transformers and converter transformers
- Early fire warning and fire extinguishing technology for transformers, cables, and energy storage equipment
- Intelligent detection technology for new types of conductors
- Intelligent technology for safety tools and equipment
- Anti-misoperation technology
- Human replacement for high-risk operation
- Major electromagnetic risk prevention and control technology for key equipment
- Operation status detection and evaluation of GIL integrated pipe gallery projects
-
- 变电站智能机器人巡检, 配电房多媒体监控, 智慧安监系统
- 变压器和换流变压器有载分接开关的健康状态检测和风险预警
- 变压器、电缆、电化学储能装备的火灾早期预警及灭火联动技术
- 新型导线智能检测技术
- 安全工器具智能化技术
- 防误操作技术
- 高危作业人工替代
- 关键设备重大电磁风险防控技术:
- GIL综合管廊工程运行状态检测与评估
-

Synergy of source-grid-load-storage with weather and carbon

6 源-网-荷-储与气-碳协同

Focusing only on the interaction and coordination of the four elements—source, grid, load, and storage is not enough

- The normalization of extreme weather has become an important factor for safe operation of the new power system, and the study of the impact of meteorology on power has become a new academic direction.

除关注源-网-荷-储四要素互动协调外，**极端气候常态化**成为影响新型电力系统安全运行的重要因素，气象对电力的影响研究成为一门新的学科方向。

- Reducing electricity carbon emissions is another important indicator for the operation of new power system. Electricity carbon factors with time-sharing and area partitioning will guide the load to shift to low-carbon time periods and regions, becoming an important consideration for the future new power system.

降低电碳排放是新型电力系统安全运行的重要指标。**分时分区电碳因子**将引导负荷向低碳时段和区域转移，成为未来新型电力系统安全运行的重要考量。



The safe operation of the new power system should focus on the interaction and coordination of the six elements: source, grid, load, storage, weather, and carbon

新型电力系统安全运行，应关注源-网-荷-储-气-碳六要素的互动和协调

Extreme weather and terrain features intensify the impact on renewable generation

极端天气和地形地貌对新能源发电影响加剧

- The meteorological conditions are complex and the terrain and landforms are diverse for large-scale wind and solar power plants in offshore, desert, high-altitude areas.
- Global climate change is intensifying, and various types of extreme weather such as dust storms, blizzards, and typhoons are becoming increasingly frequent
- 海上、沙漠、高海拔等地区大型风光电站的气象条件复杂、地形地貌多元
- 全球气候变化加剧，沙尘、暴雪、台风等各类极端天气日益频发



Complex weather conditions and diverse terrain and landforms
气象条件复杂、地形地貌多元



dust storms, blizzards and typhoons becoming increasingly frequent
沙尘、暴雪等极端天气频发

The integration of meteorology and electric power has become a new academy

气象电力 成为新的学科方向

The integration of meteorology and electric power, named meteorological power, is empowered to the entire power production, with meteorological data and technology closely link with renewable power generation forecasting, power grid operation and maintenance protection. For example:



- ◆ Accurate forecasting of renewable power generation.
- ◆ Specialized technologies for grid disaster prevention, control, operation and maintenance developed for different meteorological disasters, such as typhoon forecasting, tower risk monitoring, etc.
- ◆ An integrated decision-making platform will achieve data integration and efficient application. Such as meteorological monitoring, wind, solar and hydropower forecasting; providing the power sector with full-process services from early warning to post-disaster assessment.

气象与电力融合形成的**气象电力技术**，核心是靠**气象数据与技术**赋能电力生产全流程，涵盖新能源发电预测、电网运维防护等多个关键环节，例如：

- 新能源发电精准预测。
- 针对不同气象灾害研发的**电网灾害防控与运维专项技术**。如台风预报模型；杆塔风险监测等。
- 实现**数据整合与高效应用的一体化决策平台**。如集成气象监测、风光水电预测；为电力部门提供从预警到灾后评估全流程服务。

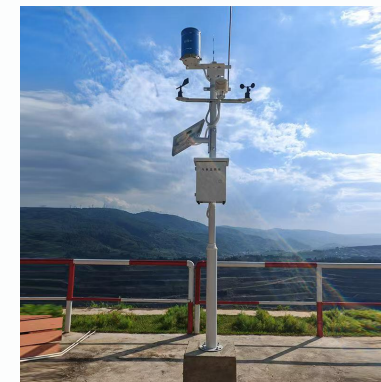
The integration of meteorology and electric power has become a new academic direction

气象电力 成为新的学科方向

Deep Coupling of Meteorology and Power - New Trends in Power Security and Emergency Response

气象与电力深度耦合 - 电力安全与应急的新趋势

- ◆ **Mini weather stations and wind-solar meteorological monitoring devices** - Refined decomposition of numerical weather prediction, providing data support for renewable power generation and ensuring station safety.
- ◆ **UAV swarm air-ground collaborative 3D modeling and image analysis**: Combining UAV aerial photography with supplementary measurements from ground 3D scanners, handheld terminals and other equipment, fusing aerial and ground data to generate 3D models.
- ◆ **Emergency plan generation based on Large Language Models (LLMs)**: Utilizing the knowledge reserve and scenario adaptation of LLMs to quickly generate standardized, detailed, and actionable emergency plans.
- ◆ **Collaborative command for power emergency repair** - Based on a satellite communication support system, achieving efficient full- management of power emergency repair through resource linkage.
- ◆ **Emergency material transportation and reserve based on extreme weather forecasting**: Matching material reserve types according to forecast levels, achieving precise supply through dynamic, and avoiding material accumulation and shortages.
- **微型气象站和风光气象监测装置** - 数值天气预报的精细化分解, 为新能源发电提供数据支持, 保障电站安全。
- **无人机群空地协同3D建模和图像解析**: 无人机空中航拍与地面三维扫描仪、手持终端等设备补测相结合, 将空中与地面数据融合, 生成3D模型。
- **基于大语言模型LLM的应急预案生成**: 利用LLM的知识储备能力与场景适配能力, 快速生成内容规范详实、可落地的应急预案。
- **电力应急抢修的协同指挥** - 基于卫星通信支持系统, 通过资源联动, 实现电力应急抢修的全流程高效管控。
- **基于极端天气预报的应急物资调运和储备**: 按预报等级匹配物资储备类型, 通过动态调度实现精准供应, 避免物资积压和短缺。



The carbon emission of electricity is an important consideration for the new power system

7 电碳排放强度是新型电力系统的重要考量

Carbon emissions in the power sector account for over 40% of the total emissions in the energy sector. To achieve the goals of "carbon neutrality", the power sector is the main battleground.

电力行业碳排放超过能源行业排放总量的40%，实现“碳达峰、碳中和”目标，电力行业是主阵地。

Figure removed

电力负荷及其分布与碳排放有密切的相关关系

The implementation of international carbon tariffs will raise the export costs of energy-intensive products, forcing countries to accelerate the green transformation of their industries and the unified construction of footprint accounting systems.

国际碳关税实施将会抬高高耗能产品出口成本，倒逼各国加速产业绿色转型和碳足迹核算体系的统一建设。

International:

EU (CBAM): Phased implementation of carbon tariffs.

UK: Implementation of its own carbon border adjustment mechanism in 202.

Other countries / International level: 196 countries reached a carbon tariff agreement at the COP30 climate summit, imposing carbon tariffs on high-emission goods such as and cement starting from 2027.

国际:

- ◆ 欧盟 (CBAM) : 分阶段推进碳关税。
- ◆ 英国: 2027 年实施本国碳边境调节机制。
- ◆ 其他国家 / 国际层面: COP30 气候峰会中 196 个国家达成碳关税协议, 2027 年起对钢铁、水泥等高排放商品征收碳关税。

The carbon emission intensity of electricity is an important consideration for the new power system

电碳排放强度是新型电力系统的重要考量



Through a precise time-and-region-based electricity carbon accounting mechanism, it is possible to link up with the international carbon tariff system, and to respond government carbon reduction policies.

- ◆ Time based: Guide the load to shift to low-carbon time periods, and "load following carbon" will become a new driver for demand response on the electricity consumption side.
- ◆ Region based: Guide energy-consuming loads to shift to low-carbon regions, and "load following carbon" will promote redistribution of regional loads

通过精准的**分时分区**电碳核算机制接轨国际碳关税体系，应对政府的减碳政策

- ◆ 分时：引导用电负荷向低碳时段转移，“荷随碳动”将成为**用电侧需求响应的新动能**
- ◆ 分区：引导耗能负荷向低碳区域转移，“荷随碳动”将促进**地域电力负荷的重新布局**

"Load following carbon" will inevitably lead to significant changes in the planning, construction, dispatch, and operation of the new power system, and will show an important impact on grid architecture and safety operation of the West-to-East Power Transmission in China.

“荷随碳动”必然引发新型电力系统的规划、建设、调度和运行等发生重大改变，对西电东送的电网架构和安全运行产生重要影响。

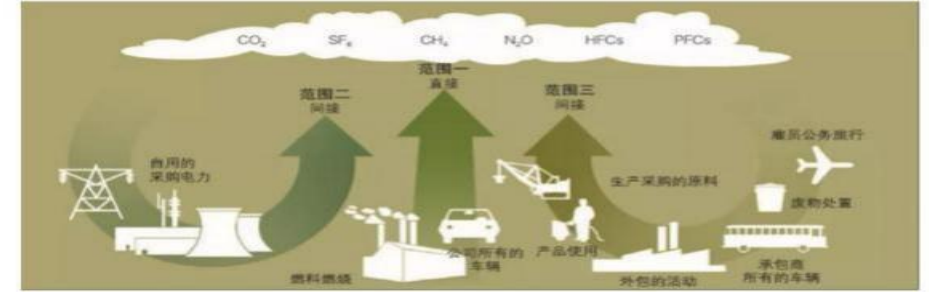
A precise and unified carbon accounting system is the foundation for promoting carbon neutrality in the new power system

精准统一的碳核算体系是推进新型电力系统碳中和的基础

Based on the characteristics of the new power system, design a reasonable carbon emission monitoring method to calculate the carbon dioxide emissions produced by the enterprises.

根据新型电力系统的特征，设计合理的碳排放监测方式，计算企业生产的二氧化碳排放量。

图 1：温室气体排放方式



资料来源：《温室气体核算体系：企业核算与报告标准》，天风证券研究所

Greenhouse Gas Emission Approaches

Calculation formula for emissions from self-purchased electricity:

Indirect CO₂ emissions from self-purchased electricity =

Self-purchased electricity consumption * Average CO₂ emission factor (of the grid area where the enterprise's production site is located in the verification year)

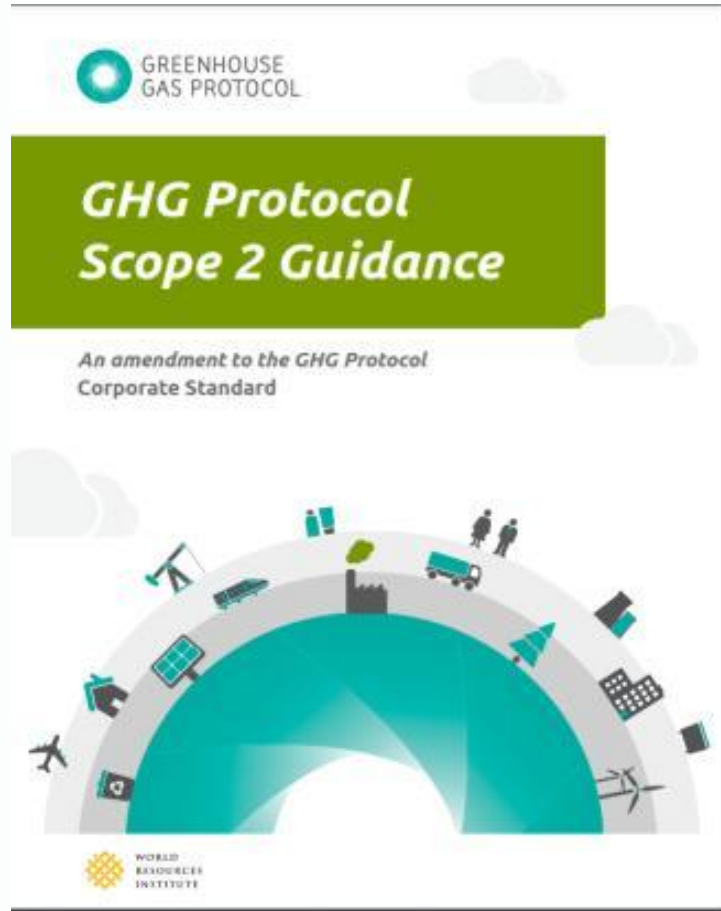
企业自购电力排放的计算公式：

自购电力所带来的CO₂间接排放 = 自购电量 * (核查年份企业生产场地所属电网区域的) CO₂平均排放因子。

US think tank proposes to revise greenhouse gas emission carbon factors

美国智库提出修改温室气体排放碳排因子

2015版：属地和市场



2015 Version: Local Presence
& Market Orientation

time

时间上:

annually

按年度

By

**quarter/month/
week/day/hour**

季度/月度/周/日/ 小时

space

空间上:

region

区域

**Region - Balance
Control Area -
Load Park to Load
Node**

区域 — 平衡控制区 — 负荷园区
到 负荷节点

U.S. Congress requires EIA to publish hourly average carbon emission factors

美国国会立法要求EIA发布小时级平均和边际碳排放因子

Sec. 40412. Data collection in the electricity sector.

Section 40412 requires the Administrator of the EIA to establish, within 90 days, an online database for the bulk power system (i.e., the electricity transmission system) in the contiguous United States. The database may build upon existing databases such as EIA's Hourly Electric Grid Monitor. Within one year, Administrator of the EIA must undertake additional activities:

- To the maximum extent practicable, add hourly operating data such as demand, demand forecasts, generation by fuel type, electricity storage and discharge, and **marginal greenhouse gas emissions rate**.
- Establish a system to harmonize its data collection with EPA and state or regional energy credit registries.^{5b}
- Establish a system for providing data broadly related to the integration of renewable energy as well as disruptions caused by cyberattacks, physical attacks, extreme weather events, or other causes.
- Establish a system for providing data on the electricity distribution system, including the delivered generation mix for each load-serving entity (i.e., utility) and the use of distributed energy resources (e.g., rooftop solar).

SEC. 40419. HARMONIZATION OF EFFORTS AND DATA.

Not later than 1 year after the date of enactment of this Act, the Administrator shall establish a system to harmonize, to the maximum extent practicable and consistent with data integrity—

(1) the data collection efforts of the Administrator, including any data collection required under this subtitle, with the data collection efforts of—

(A) the Environmental Protection Agency, as the Administrator determines to be appropriate;

(B) other relevant Federal agencies, as the Administrator determines to be appropriate; and

(C) State or regional energy credit registries, as the Administrator determines to be appropriate;

(2) the data collected under this subtitle, including the operating data on electricity generation collected under section 40412(a), with data collected by the entities described in subparagraphs (A) through (C) of paragraph (1), including any measurements of greenhouse gas and other pollutant emissions collected by the Environmental Protection Agency, as the Administrator determines to be appropriate; and

(3) the efforts of the Administrator to identify and report relevant impacts, opportunities, and patterns with respect to energy use, including the identification of community-level economic and environmental impacts required under section 40413(b)(1)(C), with the efforts of the Environmental Protection Agency and other relevant Federal agencies, as determined by the Administrator, to identify similar impacts, opportunities, and patterns.

Section 40412 of the IIJA calls on EIA to report on hourly operating data, including “where available, the estimated marginal greenhouse gas emissions per megawatt hour of electricity generated” within each balancing authority and by pricing node.⁵ Sections 40412 and 40419 of the law also call on EIA to harmonize its electric system operating data with GHG and other pollutant emissions data collected by EPA, other relevant data collected by EPA or other federal agencies, and data collected by state or regional energy credit (REC) registries. The resulting integrated data set should include net generation data and “where available, the average and marginal greenhouse gas emissions by megawatt hour of electricity generated within the boundaries of each balancing authority,” to be offered on a real-time basis through a publicly accessible application programming interface.⁶

- Accelerate the green basic system focusing on carbon accounting
- Promote the construction of the carbon sink market and implement emission reduction responsibilities
- 加快碳核算、碳账户为重点的绿色基础制度
- 推进碳汇市场建设，落实减排责任

Establishing an internationally recognized carbon footprint and carbon standard certification system 建立国际互认碳足迹、碳标准认证体系

实现碳中和是建设新型电力系统的战略目标，通过电碳足迹溯源助力政府精准推进减碳战略，引导电力用户低碳零碳运行。

Achieving carbon neutrality is a strategic goal for building a new power system. By tracing the carbon footprint of electricity, it helps the government accurately advance carbon reduction strategies and guides power users toward low-carbon or zero-carbon operations.



- ◆ 深化新能源、电工装备、节能环保等领域国际标准研制。
- ◆ Strengthen international standards research and development in the fields of renewable energy, electrical equipment, energy saving, environmental protection, etc.
- ◆ 健全碳排放核算、核查、披露和认证制度体系，建立产品全生命周期碳足迹标准和标识制度。
- ◆ Accomplish accounting, verification, disclosure and certification systems for carbon emissions.
- ◆ 培育与国际接轨的碳排放认证检测服务机构。
- ◆ Cultivate carbon emission consultation organizations with international standards.

建立国际互认碳足迹、碳标准认证体系，创建公平公正的国际碳足迹管理秩序，**是在新型电力系统建设中保障能源环境安全的重要举措。**

Establishing an internationally recognized carbon footprint and carbon standard certification system to create a fair and equitable international carbon footprint management order is a crucial measure for ensuring energy and environmental security in the construction of a new power system.



Constructing a single electricity-carbon map based on power big data

基于电力大数据构建电碳一张图

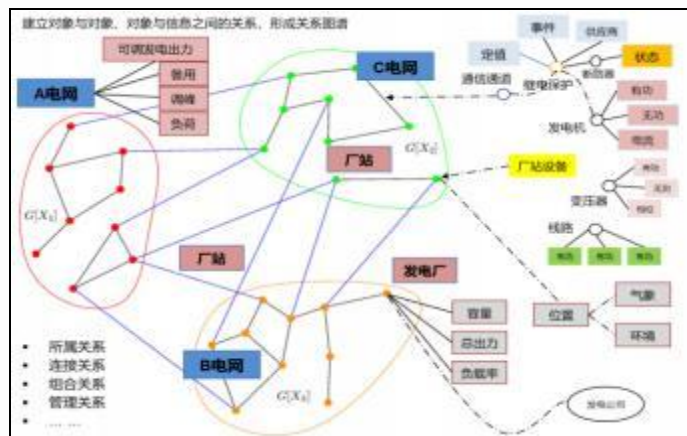
The "Electricity-Carbon Map" is an electricity-carbon coupling model constructed using graph computing based on power emissions and related data from power generation and user sides. It enables carbon flow tracking and precise, dynamic, time-varying, and region-specific quantification of electricity-carbon factors, to support government to establish a carbon emission accounting system, help enterprises reduce energy consumption, and promote development of the green electricity trading market.

电碳一张图是基于电源侧、电网侧、用户侧电力碳排放及相关数据，采用图计算构建的全景电-碳耦合模型，实现碳流追踪和精准、动态的分时、分区电碳因子量化，支撑政府建立电力碳排放核算体系，助力企业降低能耗，促进绿电交易市场发展。



Real power grid

真实电网



Electricity-carbon network constructed using graph models

使用图模型构建的电-碳网



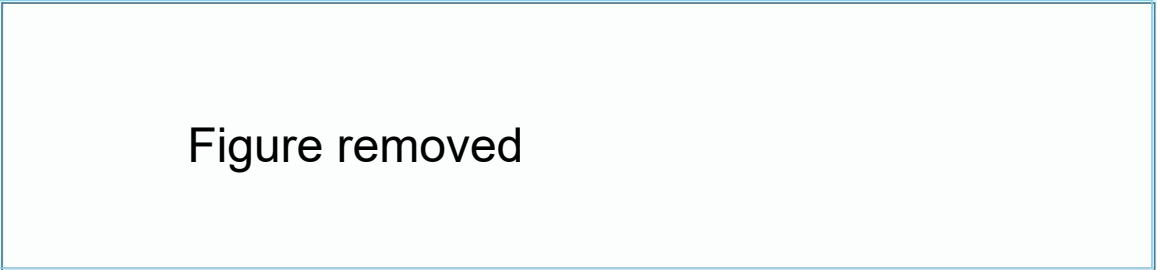
Carbon flow tracking and power carbon intensity calculation

碳流追踪和电力碳强度计算

Online calculation and analysis of carbon emission intensity and carbon footprint tracking

碳排放强度在线计算分析与碳足迹追踪

Develop a graph-technology-based data foundation and key application software for the "Electricity-Carbon Integrated Map" to track electricity-carbon footprints, enabling the calculation of power carbon intensity and carbon tracking across multiple dimensions, and achieving the visualization of carbon flows, electrical energy flows, and other elements on "Electricity-Carbon Integrated Map".



Prototype
demonstration

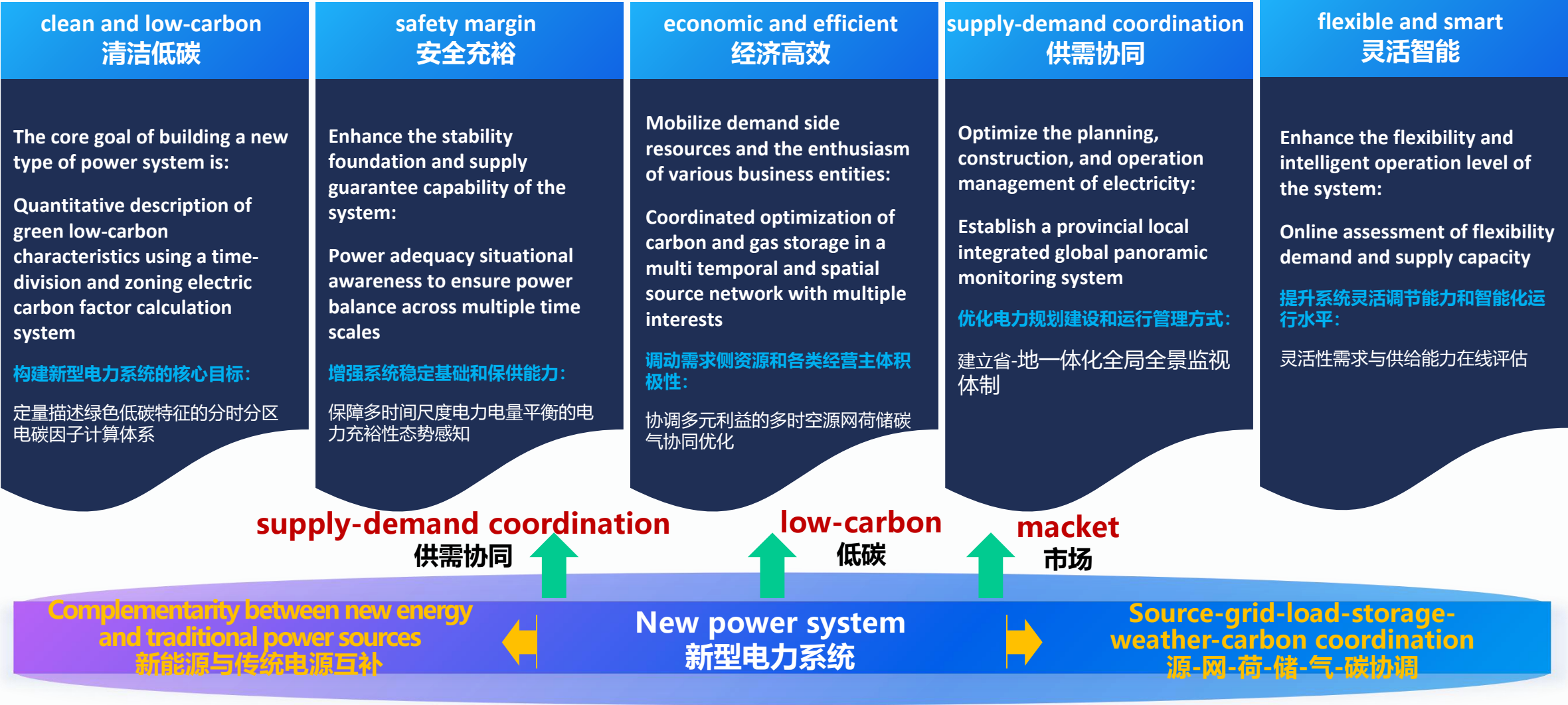
calculation
and analysis

开发基于图技术的电碳一张图时空数据底座及电碳迹追踪关键**应用软件**，实现多时空维度的电力碳强度计算及碳轨迹追踪，实现碳流、电能流等在电碳一张图上的时空展示。

Model
building

Optimization of six elements of source, grid, load, storage, weather and carbon to support safe operation of new power system

源网荷储气碳六要素协同优化, 支撑新型电力系统安全运行



Features
功能亮点

Source load prediction 源荷预测	Carbon footprint management 碳迹管理	Situation Awareness 态势感知	Panoramic monitoring 全景监控	Collaborative optimization 协同优化	Flexibility optimization 灵活性优化
Meteorological forecast and warning 气象预测预警	Real time calculation of electric carbon factor 电碳因子实时计算	State estimation 状态估计	Source resource monitoring 电源资源监测	Section monitoring warning 断面监测预警	Flexibility Needs Assessment 灵活性需求评估
Full time and space power prediction 全时空功率预测	Online prediction of electric carbon factor 电碳因子在线预测	Power flow calculation 潮流计算	Grid resource monitoring 电网资源监测	Security calibration scheduling 安全校正调度	Flexible supply assessment 灵活性供给评估
Multi time and space load forecasting 多时空负荷预测	Carbon flow trajectory tracking 碳流轨迹追踪	Static security analysis 静态安全分析	Load resource monitoring 负荷资源监测	Source-Net-Load-Storage-Carbon Wether collaborative optimization 源网荷储碳气协同优化	Hierarchical partition aggregation 分层分区聚合
Load characteristic analysis 负荷特性分析	Carbon asset management 碳资产管理	Short circuit current & transient analysis 短路电流&暂态分析	Energy storage resource monitoring 储能资源监测		

技术底座

Technology Base



AI empowerment



AI
赋能

Knowledge graph 知识图谱	Retrieval Enhanced Generation (REG) 检索增强生成 (REG)	Large model for power dispatching 电力调度大模型	Predictive large model 预测大模型	Meteorological Large Model 气象大模型	Multi-agent 多智能体
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Data
多源数据

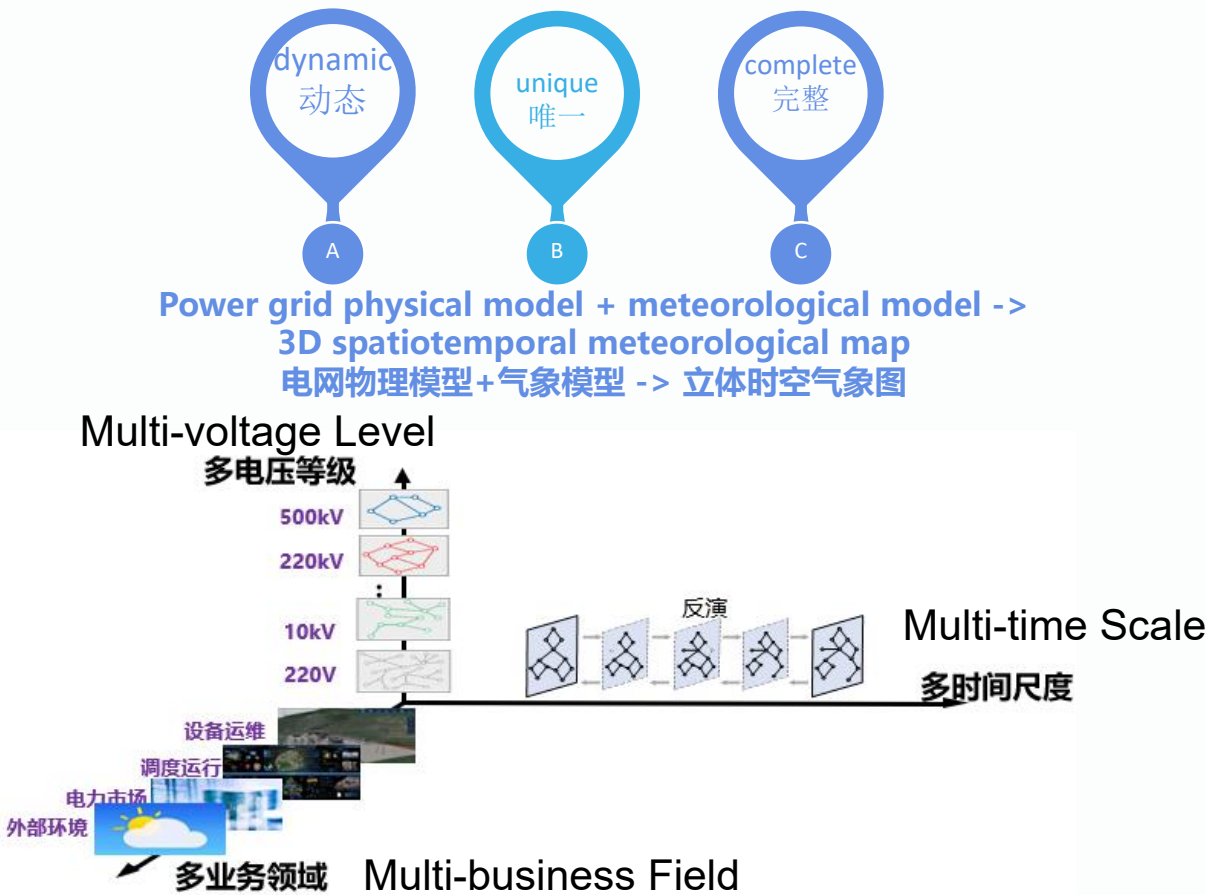
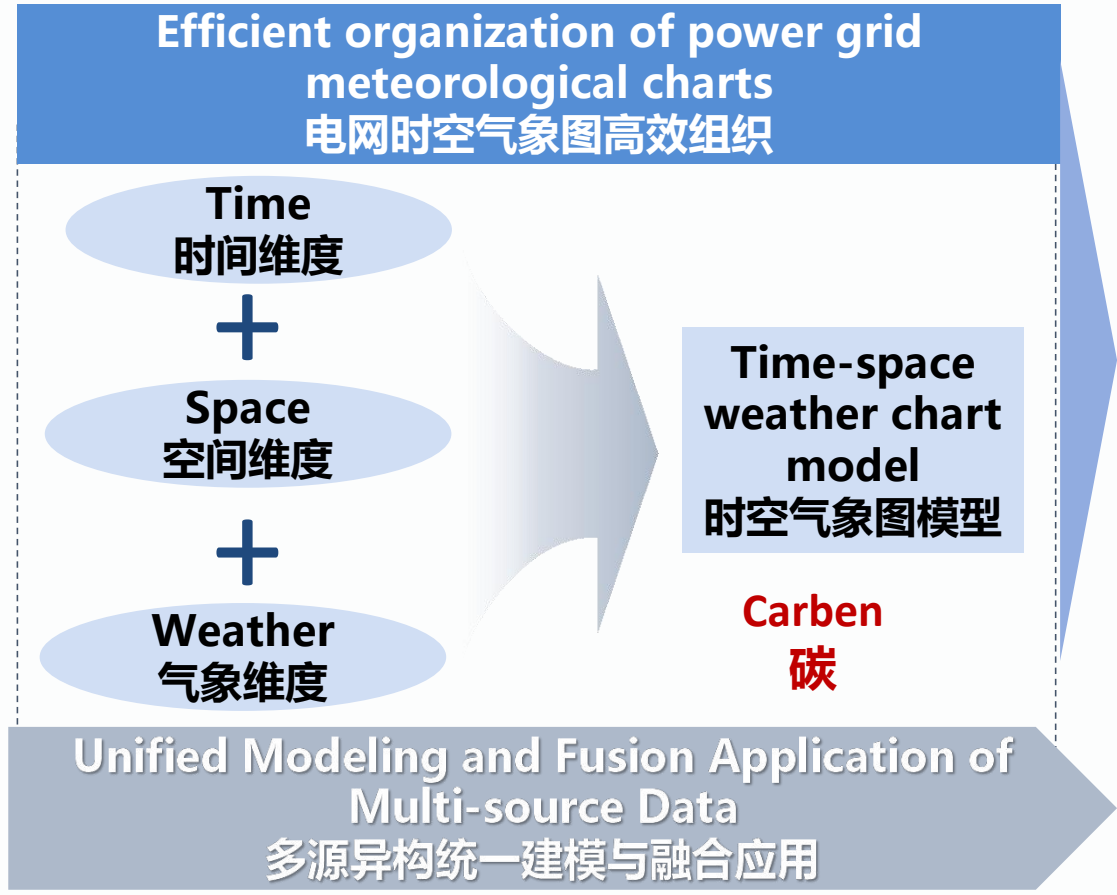
 CIM power grid topology data CIM电网拓扑数据	 Scheduling automation data 调度自动化数据	 New energy centralized control system 新能源集控系统	 Smart IoT data 智能物联网数据	 User side electricity consumption data 用户侧用电数据	 meteorological monitoring data 气象监测数据
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Construction of a spatiotemporal data model for power, carbon, and gas in the power grid on a single map

电网时空电碳气一张图数据模型构建

Achieved the dynamic complete mapping of the time-space graph model for 10kV to 500kV power systems, overcame the challenges of modeling and fusion application for multi-source power grid data, and supported the unified modeling and analysis of large-scale power systems.

实现了10kV~500kV电力系统时空演化图模型的动态完备映射，攻克电网多源数据统一建模与融合应用难题，支撑大规模电力系统多时态统一建模与分析。



Key technologies of the collaborative optimization system for source, grid, load, storage, weather and carbon

源网荷储气碳协同优化系统的关键技术

Graph technology foundation 图技术底座

- Physics-Carbon Flow Coupled Modeling: **Network Model for Power Grid Topology and Dynamic Carbon Flow Based on Graph Database**
- Multi-source Dynamic Interconnection: **Rapid Topology Analysis of Ten-Millions of Device Nodes, and Real-time Interconnection of Ten-Millions of Source-Grid-Load-Storage Resource Devices**
- **物理-碳流耦合建模**: 基于图数据库构建电网拓扑与动态碳流的网络模型
- **多源异构动态互联**: 支持千万级设备节点快速拓扑分析, 千万级源-网-荷-储资源设备的实时互联

AI-enhanced decision making AI增强决策

- "Large Model Knowledge Graph" Dual Engine: **Build a power dispatch large model to support traceability of complex decision-making**
- Intelligent Game Deduction: **Operations optimization+reinforcement learning+ game theory to intelligently deduce games such as energy trading under multi-objectives.**
- **“大模型+知识图谱”双引擎**: 构建电力调度大模型; 基于“设备-能源-碳排-市场”多维知识图谱, 支撑复杂决策溯源
- **智能博弈推演**: 运筹优化+强化学习+博弈论, 智能推演多目标下能源协同、跨系统能源交易等博弈

Electric-carbon coupling analysis 电碳耦合分析

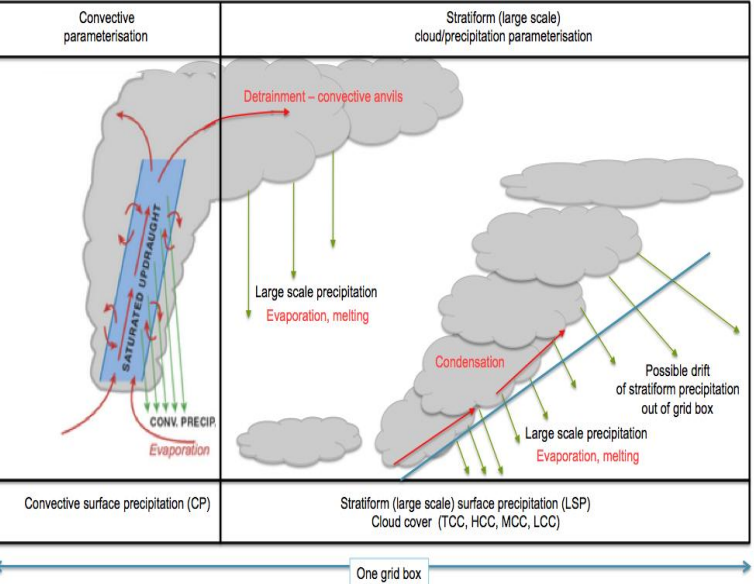
- Dynamic Integration of Electricity Carbon Factors: **Integrate 20+ carbon accounting authorities such as IPCC, China's Ministry of Ecology and Environment, and EUE TL**
- Real-time Tracking and Allocation of Carbon Flows: **Achieving "per-kWh" carbon emission traceability based on blockchain technology**
- **电碳因子动态集成**: 整合IPCC、中国生态环境部、欧盟ETL等20+权威碳核算标准
- **碳流实时追踪与分摊**: 基于区块链技术实现“度电级”碳排追溯

Comprehensive and accurate weather, power generation, and load forecasting

全面精准的气象、发电及负荷预测

- Carry out more comprehensive and accurate meteorological, power generation, and load forecasting, to mitigate system risks under the impact of fluctuations of renewable energies.
- 开展更全面、精准的气象、发电及负荷预测，实现新能源时序波动影响下的系统风险平抑。

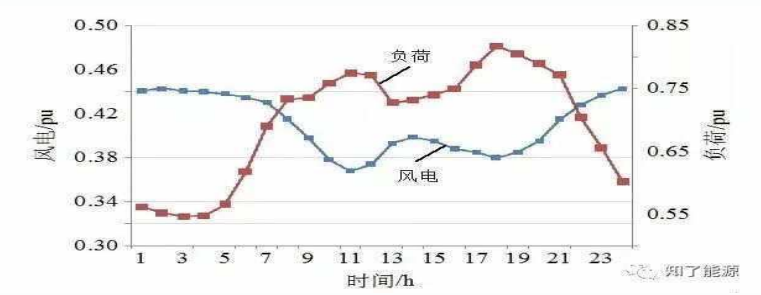
Large-scale extreme weather events 大规模的极端气象事件



Threats to power generation and transmission equipment 对发电、输电设备的威胁



Impact on power generation and consumption characteristics 对发电用电特性影响

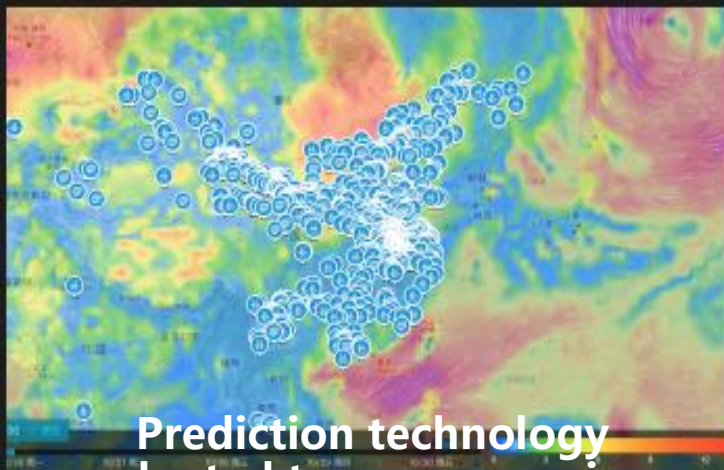


Climate change has a significant impact on the operation of renewable energies, bringing numerous effects to the power grid and equipment, which requires continuous improvement in the scope and accuracy of forecasting.

气候变化对新能源运行影响明显，对电网、设备带来诸多影响，需要持续提升预测范围和精度

Fused meteorological data, AI-based precise prediction

融合气象数据，基于AI的精准预测



Prediction technology adapted to new scenarios

预测技术适配新场景

- Rolling forecast for the next 72 hours every 15 minutes
- Real time access to the latest data for ultra short term calculations
- Short term forecast updated 4 times a day
- Probability forecast results for P40 and P70 in the next 72 hours



Establish an evaluation index system

建立评价指标体系

- Establishing prediction targets in new scenarios
- Establishing evaluation targets in new scenarios
- Establish accuracy evaluation indicators for small wind sections



Accuracy enhancement scheme

准确率提升方案

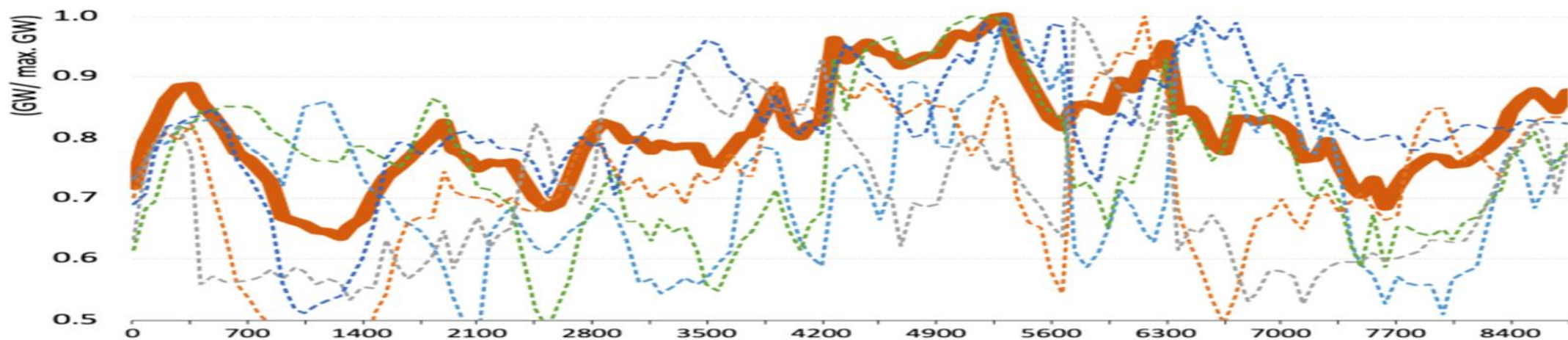
- Optimization of meteorological sources based on EC ensemble forecasting
- New Algorithm for Power Limitation and Small Wind Section Scenarios
- Real time data acquisition and analysis with higher resolution

Online security assessment of power systems with source-grid-load-storage-weather-carbon synergy

源网荷储气碳协同的电力系统安全性在线评估

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- Balance the demand and supply of power system over a larger scope and shorter time scale, integrate different output and demand characteristics, to effectively enhance the mutual aid function in power grid, and reduce the demand for new power generation and backup resources.
- Online assessment of power grid security adequacy, main equipment availability coefficient, system inertia coefficient, etc.
- 更大范围、更短尺度平衡电力系统的需求与供给，集合不同的出力与需求特性，有效发挥电网互济功能，降低对新增发电与备用资源的需求。
- 在线评估电网安全充裕度、主设备可用系数、系统惯性系数等。



Online evaluation of power guarantee adequacy, power grid safety adequacy, and power system inertia coefficient

电力保障充裕度、电网安全充裕度、电力系统惯性系数在线评估

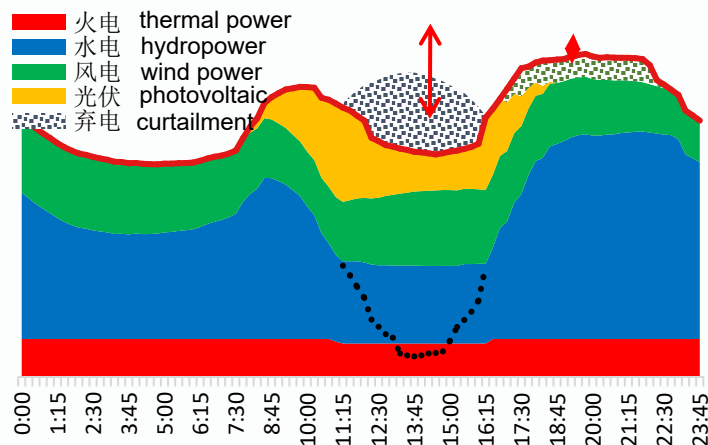
Analysis of Grid Adequacy and Flexibility Based on Source-Grid-Load-Storage-weather-Carbon Synergy

源网荷储气碳协同的电网充裕性与灵活性分析

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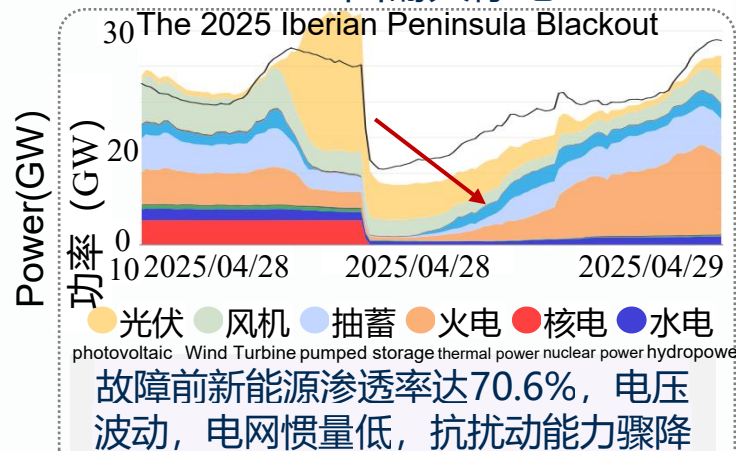
- "Abandon and shortage of renewables" has become the new normal, as extreme weather conditions such as prolonged periods of no wind and no sunlight frequently. The balance of the power system faces challenges, requiring rapid calculation of power supply adequacy and flexibility.
- Insufficient regulation capacity has become the main contradiction, with the inertia and support levels of the power grid system declined, and the risk of large-scale power outages intensifying, requiring an accurate assessment of the adjustable capacity of source, grid and load.
- “缺电与弃电并存”成为新常态，长时间无风无光等极端天气时有发生，电力系统平衡面临挑战，需快速计算**供电充裕性和灵活性能力**。
- **调节能力不足**成为主要矛盾，电网系统惯量、调节支撑水平下降，大面积停电风险加剧，需准确评估源网荷的可调节能力。

Power shortages and power curtailment coexist 缺电与弃电并存



Extreme disaster threats 极端灾害威胁

2025西葡大停电



Balance+stability risks intensify 平衡+稳定风险加剧

2025 Spain Portugal Power Outage

Before fault, the penetration rate of new energy reached 70.6%, with voltage fluctuations, low grid inertia, and a sudden drop in anti-interference ability

Improve regulation capacity, solve the coexistence of power shortages and power curtailment, and mitigate the system balancing risks under extreme weather.

提高调节能力，解决缺电与弃电并存问题，减缓极端天气系统平衡风险

Synergistic optimization of source, grid, load, storage, weather and carbon to ensure the security of the new power system

源网荷储气碳协同优化，确保新型电力系统安全

Integrate multi-source system data including day-ahead and intra-day power generation plans, new energy power and load forecasting, and virtual power plants, to build a integrated optimization platform in provincial dispatching center for source, grid, load, storage, weather and carbon.

集成日前发电计划、日内发电计划、新能源功率预测与负荷预测、负荷管理（虚拟电厂）等多源系统数据，构建省级调度源网荷储气碳一体化协同优化平台。

- Integrated modeling of source-grid-load-storage
- Carbon situational awareness of source-grid-load-storage
- Integrated weather-carbon optimization with source-grid-load-storage
- Online flexibility assessment of source-grid-load-storage
- Online calculation of time-and-region-specific electricity-carbon factors
- 源网荷储一体化建模
- 源网荷储碳态势感知
- 源网荷储气碳一体化优化
- 源网荷储灵活性在线评估
- 分时分区电碳因子在线计算





THANKS

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